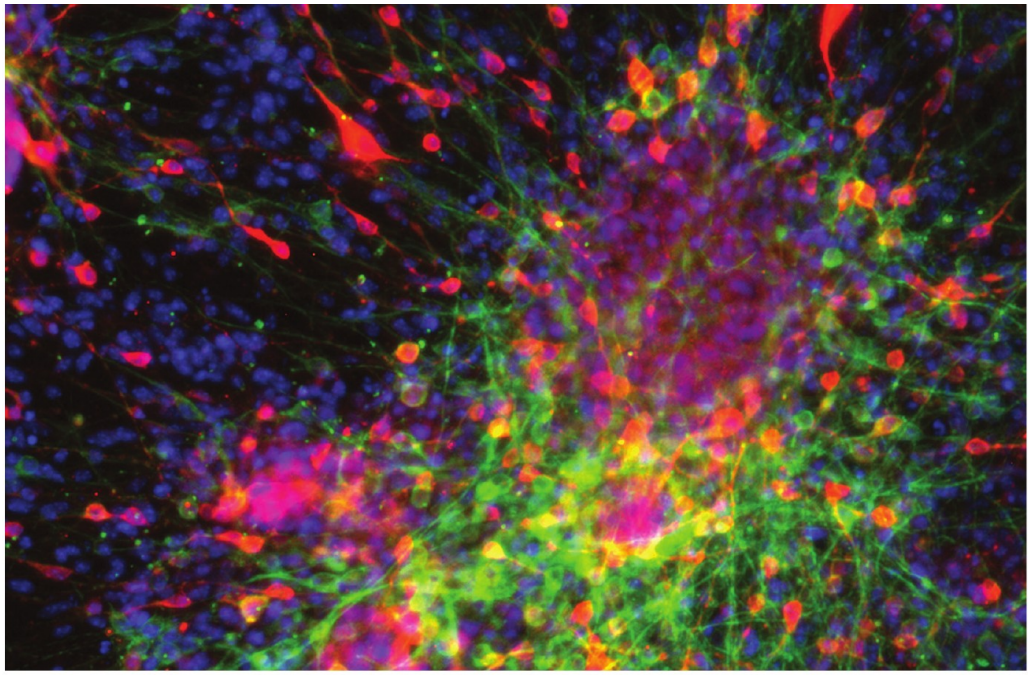


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BEHAVIORAL NEUROSCIENCE 392

Neurological Disorders

When Neural Systems Fail: Injury, Degeneration, and Infection

Fall 2026

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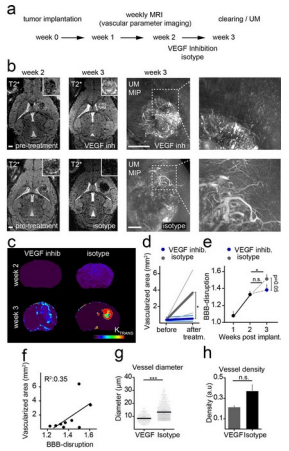
Carlson and Birkett (2022), Chapter 16 opener, p.512. Image: NIGMS/CIRM/Yaping Sun/Su Guo Lab/University of California, San Francisco/Science Source.

Tumors disturb circuits before they destroy neurons

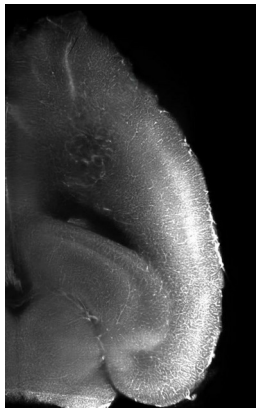
A brain tumor is a growing mass of abnormal cells. Its effects depend on where it grows as well as how rapidly its cells divide. Pressure on nearby tissue can change movement, sensation, memory, or electrical activity. The skull limits the space available for expansion, so a growing mass can displace surrounding tissue.

A benign tumor usually has a clearer boundary and does not invade surrounding tissue in the same way as a malignant tumor. Inside the skull, however, even a slowly growing mass can compress essential pathways or obstruct cerebrospinal fluid.

Glioma tissue grows beside functioning cortex. Cleared-tissue imaging of an implanted mouse tumor shows vessels at this boundary and in the surrounding brain. A fluorescent vessel label makes their branching visible as the recording passes through successive tissue depths.



Breckwoldt et al. (2016), Fig. 4.



Glioma boundary vessels. Breckwoldt et al. (2016).

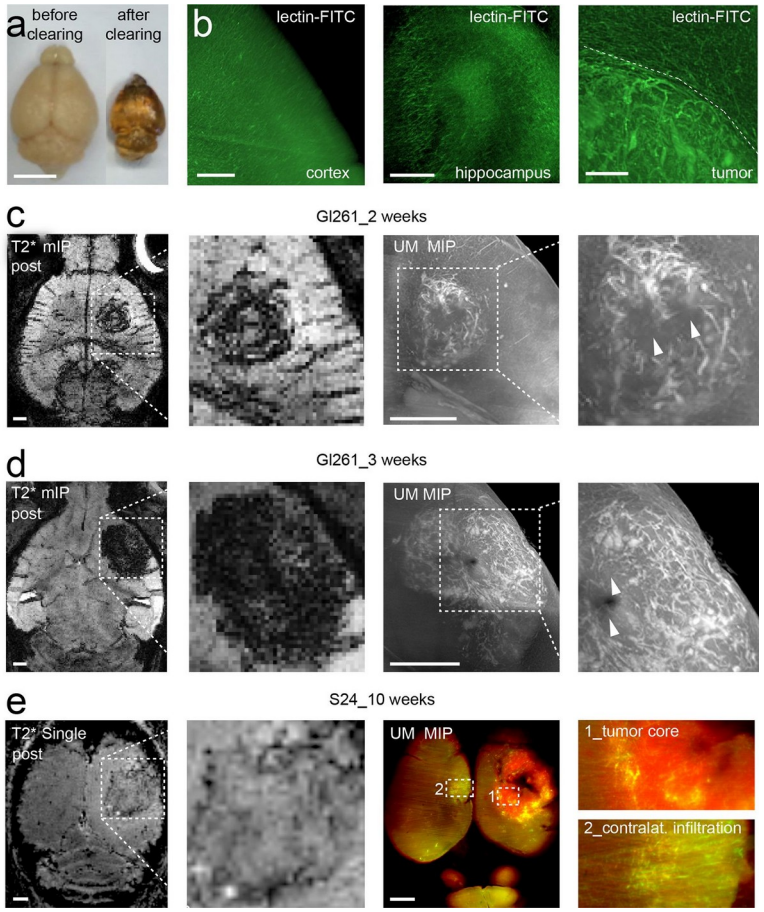
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Glioma growth requires an expanding blood supply

Glioblastoma cells require oxygen and nutrients as they multiply. They release signals that encourage nearby blood vessels to sprout. This formation of new vessels is called angiogenesis, and it changes both the tumor and the tissue around it.

In a mouse glioma study, researchers followed implanted tumors with magnetic resonance imaging. They examined the same tumors again with light-sheet imaging after clearing the brain tissue, which makes deeper structures visible without cutting the entire brain into sections.

The two imaging methods resolve different features. Magnetic resonance imaging follows vascular changes in a living animal over time. Cleared-tissue microscopy resolves smaller branches and the arrangement of vessels around invading tumor cells after the final measurement.



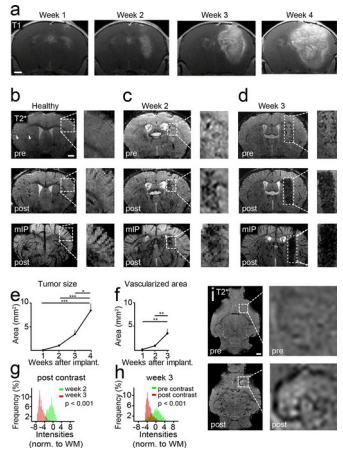
Breckwoldt et al. (2016), Fig. 2.

Contrast imaging distinguishes tumor vessels

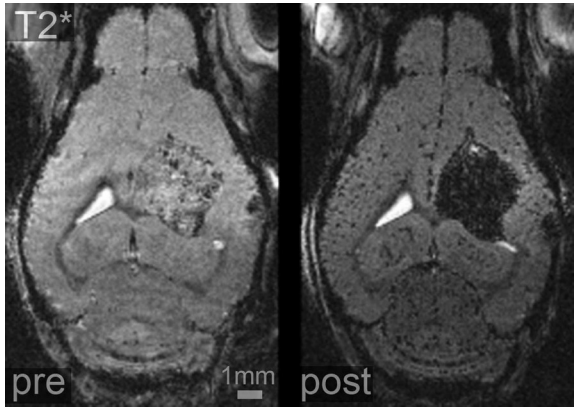
A contrast agent changes how structures appear in a magnetic resonance image. In this glioma study, images collected before and after contrast administration exposed tubular structures within the tumor. The researchers followed these structures as the tumor developed.

The vascular area increased as the implanted tumors grew. Tortuous vessels and additional branches appeared in the tumor region. The images therefore measured changes in the supply network, rather than treating tumor volume as the only sign of progression.

Vessel visibility also depends on the imaging sequence and contrast conditions. Comparing scans acquired under matched conditions allowed the researchers to follow changes in vascular structure instead of mistaking a change in acquisition conditions for new vessel growth.



Breckwoldt et al. (2016), Fig. 1.



MRI before and after contrast. Breckwoldt et al. (2016).

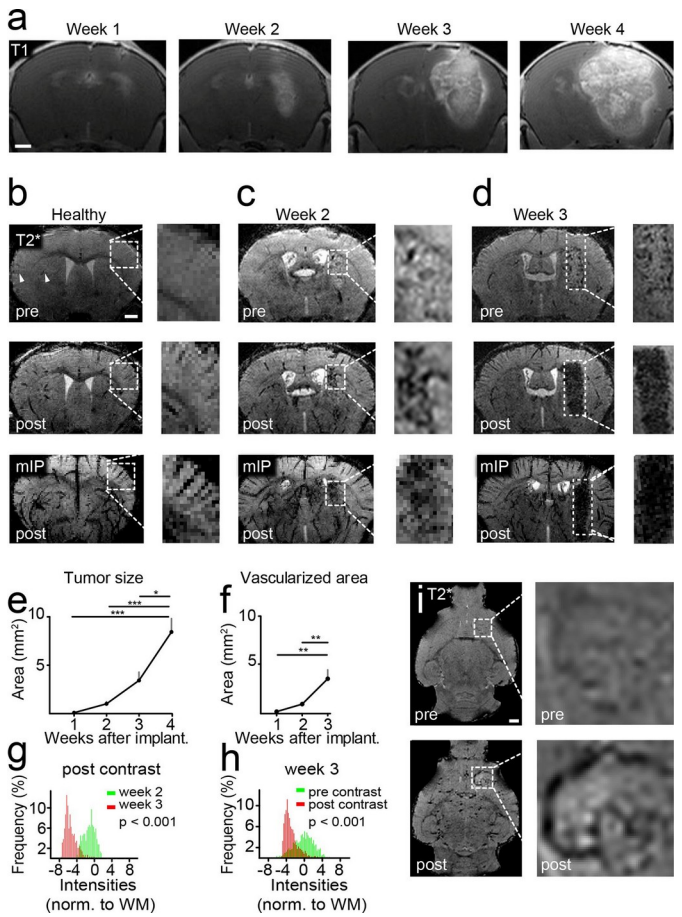
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Treatment targets both tumor cells and their access

Surgery can remove a localized mass, but infiltrating cells may extend beyond its visible boundary. Radiation damages cellular DNA, and chemotherapy interferes with processes needed for cell survival or division. These interventions act on different features of the disease.

The blood–brain barrier limits entry of many substances from blood into brain tissue. Drug delivery therefore depends on whether a compound crosses this barrier and reaches tumor cells at an effective concentration. A drug that kills cultured cells may not reach an intact tumor.

Treatment planning also protects functioning tissue. Removing a mass near a movement or language area can carry different consequences from removing a similarly sized mass elsewhere. Imaging helps locate the lesion and the pathways that need to be preserved.



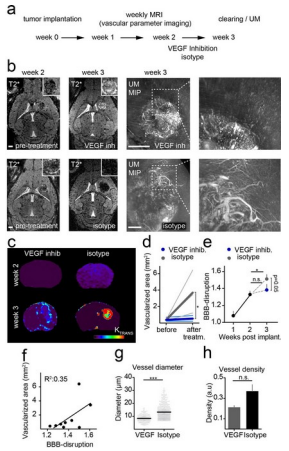
Breckwoldt et al. (2016), Fig. 1.

Blocking VEGF changes the tumor vasculature

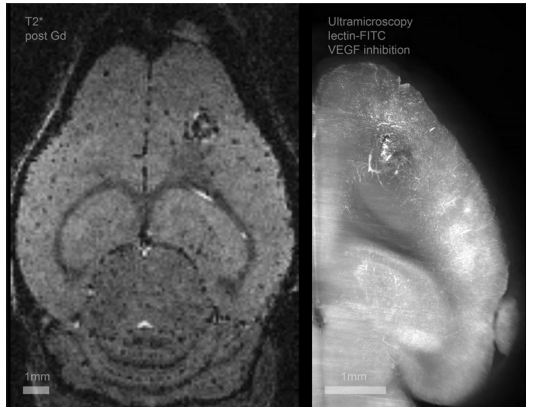
Vascular endothelial growth factor, or VEGF, stimulates blood-vessel growth. The glioma researchers inhibited this signal after tumors had formed. They then compared vascular measurements before and after treatment, using both magnetic resonance imaging and cleared-tissue microscopy.

VEGF inhibition reduced vessel caliber, vessel density, and vascular permeability in this model. Permeability describes how readily material leaves a vessel and enters surrounding tissue. Less leakage therefore represents a change in the vessel wall, alongside changes in vessel structure.

These measurements describe partial vascular normalization. The vessels became less abnormal by the measured criteria. Reduced permeability and reduced tumor size are different outcomes, so the study assessed the vascular network rather than assuming that a less leaky vessel meant every tumor cell had disappeared.



Breckwoldt et al. (2016), Fig. 4.



VEGF-inhibited tumor: MRI and microscopy. Breckwoldt et al. (2016).

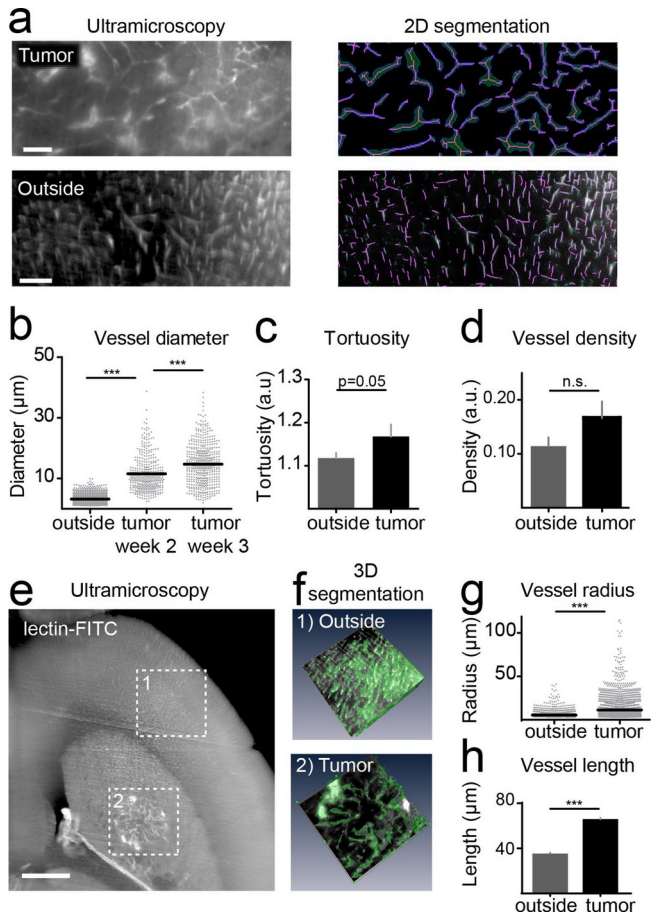
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Cleared brains reveal where glioma vessels branch

Light-sheet microscopy illuminates a thin plane within a transparent specimen. Moving through successive planes produces a three-dimensional view of labeled vessels. The glioma investigators used this approach to examine vascular organization throughout the cleared mouse brain.

Fluorescent labeling made microvessels visible against surrounding tissue. The resulting image stacks showed branching, vessel caliber, and the spatial arrangement of tumor-associated vessels. Those features could be compared with structures identified earlier in the animal's magnetic resonance images.

Matching the two datasets required locating the same tumor region in both preparations. This preserved the anatomical comparison between a living-brain measurement and a later tissue measurement, while keeping their different resolutions and acquisition conditions explicit.



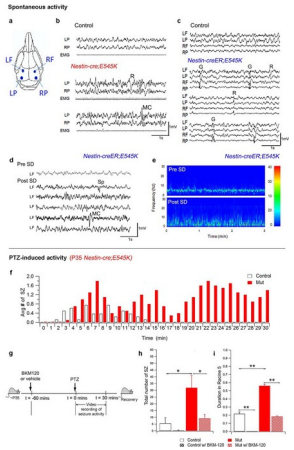
Breckwoldt et al. (2016), Fig. 3.

Seizures arise when neural activity spreads abnormally

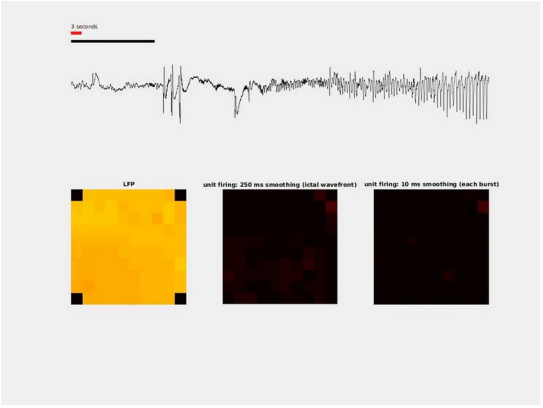
A seizure is a transient episode of abnormal, excessive or unusually synchronized neural activity. Its behavioral effects depend on the networks involved. Motor activity, altered sensation, a change in awareness, or a brief interruption of behavior can occur.

A focal seizure begins within a limited network. Activity may remain localized or spread into wider networks, including both hemispheres. A generalized seizure involves distributed networks from the outset, so the starting pattern differs from a focal event that later spreads.

Intracranial microelectrodes can record seizure activity directly from cortical tissue. In the human recording, brief discharges travel before a slower front recruits additional neurons into the seizure. Voltage and firing maps reveal different aspects of this spreading activity.



Roy et al. (2015), Fig. 7.



Human cortical discharges and seizure recruitment. Smith et al. (2022).

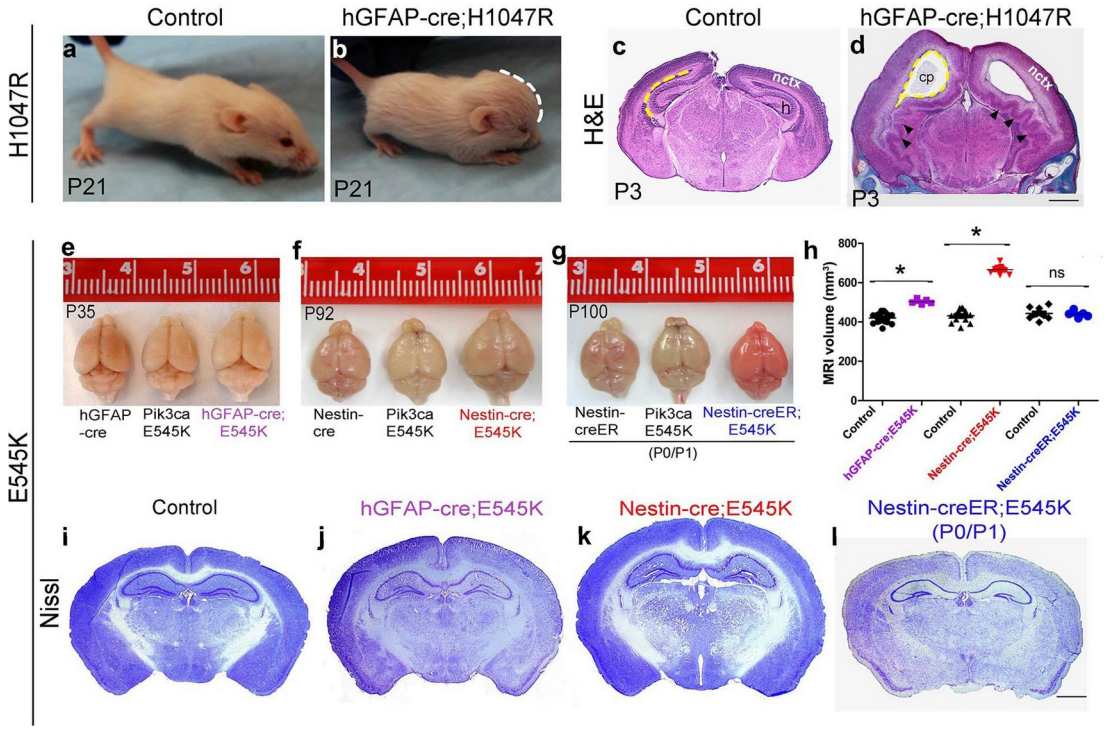
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An activating growth mutation can also produce seizures

PIK3CA encodes part of an enzyme that participates in signaling controlling cellular growth. Some activating mutations increase this signaling. In people, such mutations can occur in only a subset of cells and produce localized or extensive brain overgrowth.

Researchers introduced activating Pik3ca mutations into developing neural progenitors in mice. A progenitor is a dividing cell that produces neurons or glia. The animals developed combinations of brain enlargement, cortical malformation, altered white matter, and epilepsy.

The mutation's identity and the time at which it became active influenced severity. An early developmental change can affect many descendants of a progenitor, whereas a later change acts within a different population and stage of tissue organization.



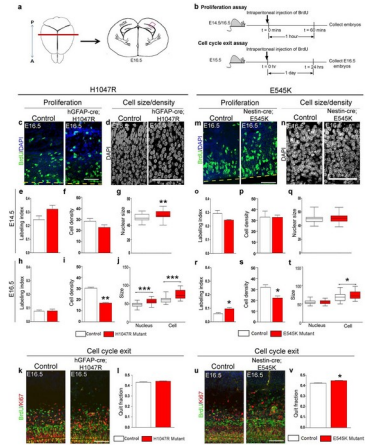
Roy et al. (2015), Fig. 1.

Overgrowth changes both cell number and cellular size

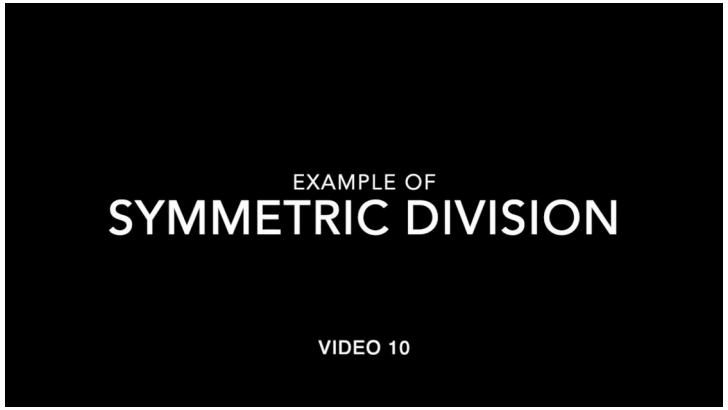
The Pik3ca models allowed researchers to examine why the brain became enlarged. Increased proliferation produced additional cells, and increased cell size expanded the tissue further. These are distinct cellular contributions to a similar visible anatomical outcome.

The cerebral cortex also developed abnormal organization. Cortical dysplasia means that the tissue's cellular arrangement is atypical, rather than simply larger. Such structural differences can change the routes through which excitatory and inhibitory neurons influence one another.

Normal neural stem cells can also increase cell number. Separate imaging of adult mouse hippocampus follows a stem cell dividing into two daughters that retain radial processes. These symmetric divisions are normal stem-cell behavior, unlike the abnormal cortical growth in the mutant mice.



Roy et al. (2015), Fig. 3.



Symmetric neural stem-cell divisions. Guo et al. (2022).

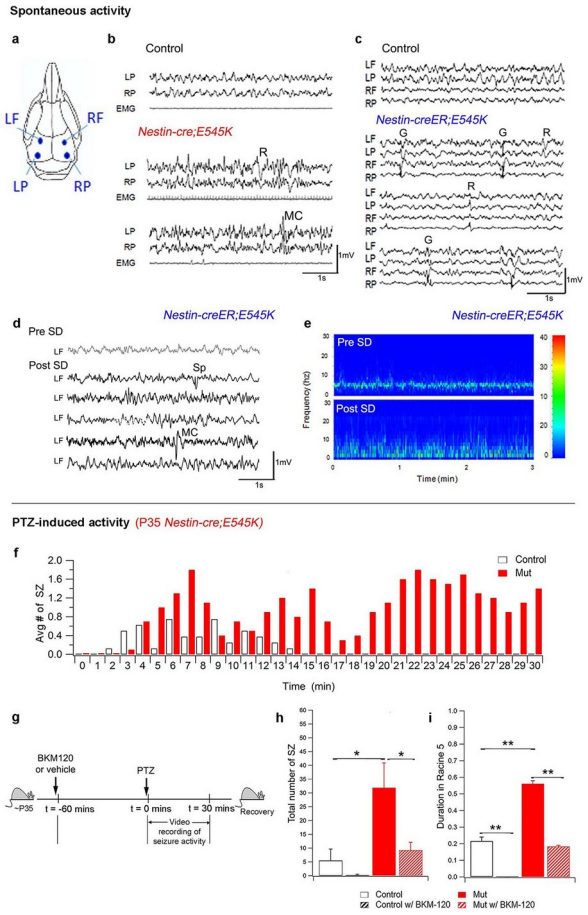
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Seizure control depends on the active network

Excitatory synapses tend to bring a neuron closer to firing, whereas inhibitory synapses can reduce firing or constrain its timing. Seizures can emerge when cellular properties and network organization allow excitation to recruit neighboring neurons repeatedly.

Antiseizure medicines act through several mechanisms, including changes in ion-channel function and synaptic transmission. The treatment chosen depends on seizure type and the person's broader clinical circumstances. A drug effective for one seizure pattern may be unsuitable for another.

Some focal epilepsies can be treated by removing or disconnecting tissue that initiates seizures. This requires identifying the seizure-generating region and assessing the functions it serves. Electrical recordings and anatomical imaging therefore answer complementary clinical questions.



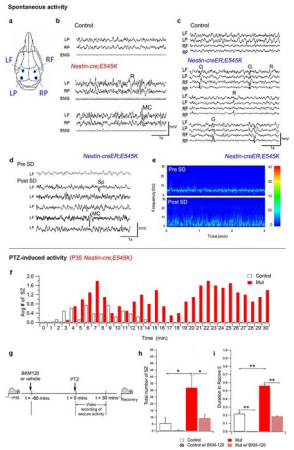
Roy et al. (2015), Fig. 7.

Pathway inhibition suppresses seizures in mutant mice

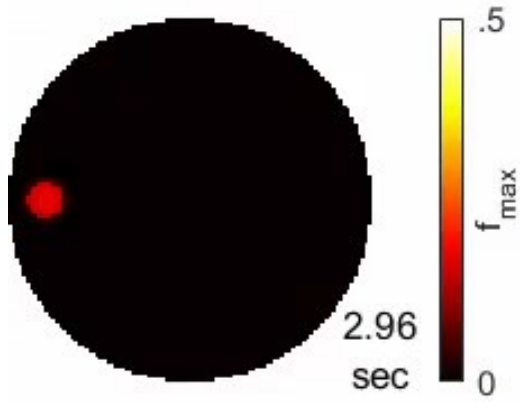
The researchers administered a PI3K-pathway inhibitor to mice with established cortical abnormalities. The intervention reduced pathway activity acutely. It did not reconstruct the developing cortex or remove the mutation from cells already carrying it.

Seizure activity decreased despite the continued presence of dysplasia. The result separates an ongoing biochemical contribution from the pre-existing anatomical abnormality. Both belong to the same disease model, but they operate on different timescales.

A separate cortical model represents recurrent excitation, inhibition, and firing adaptation, a reduction in firing after sustained activity. A brief focal input initiates a seizure that spreads through the simulated sheet. This model tests network propagation rather than the growth-pathway inhibitor administered to the mutant mice.



Roy et al. (2015), Fig. 7.



Simulated focal seizure propagation. Liou et al. (2020).

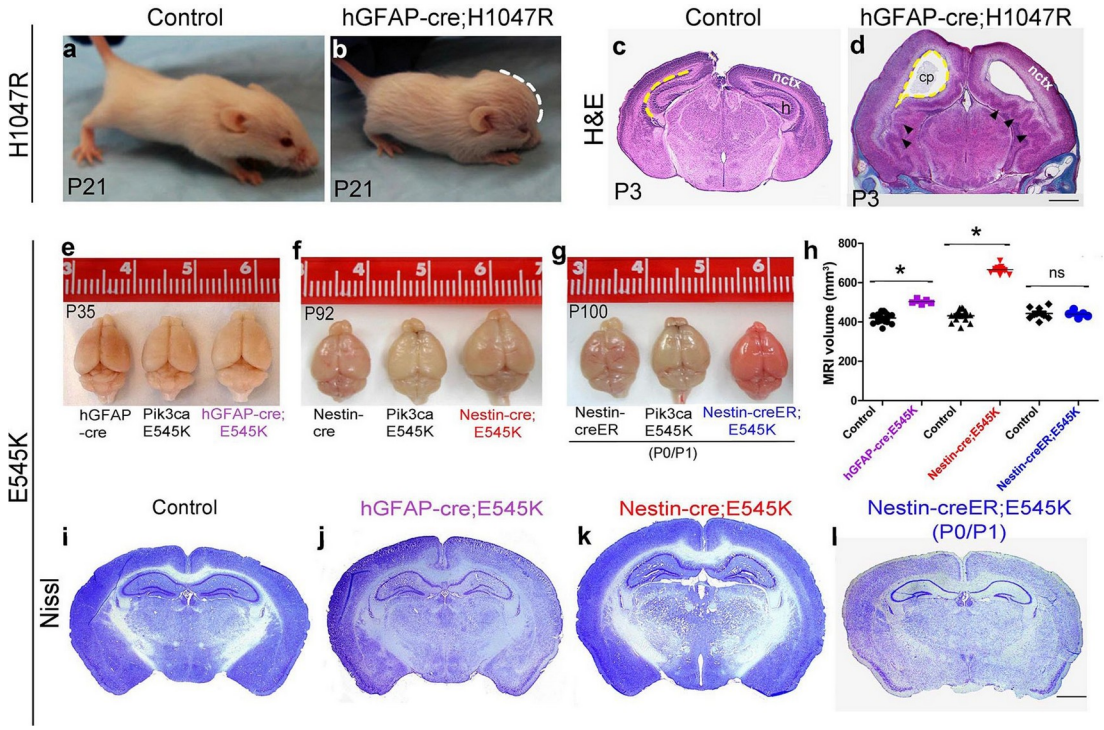
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Mutation timing changes the tissue that enters the network

A mutation activated during development can affect cells before they take their final positions or form mature connections. In the Pik3ca study, the investigators varied the developmental context of mutation expression and compared the resulting brain abnormalities.

Earlier and later expression did not produce identical phenotypes. The number and identity of affected cells, their growth, and the surrounding tissue's development changed together. The eventual seizure phenotype therefore emerged from a specific developmental history.

The acutely effective inhibitor acted on signaling that persisted in those cells. Developmental origin and ongoing physiological activity were both important experimentally: the first produced the abnormal substrate, and the second remained capable of changing seizure expression.



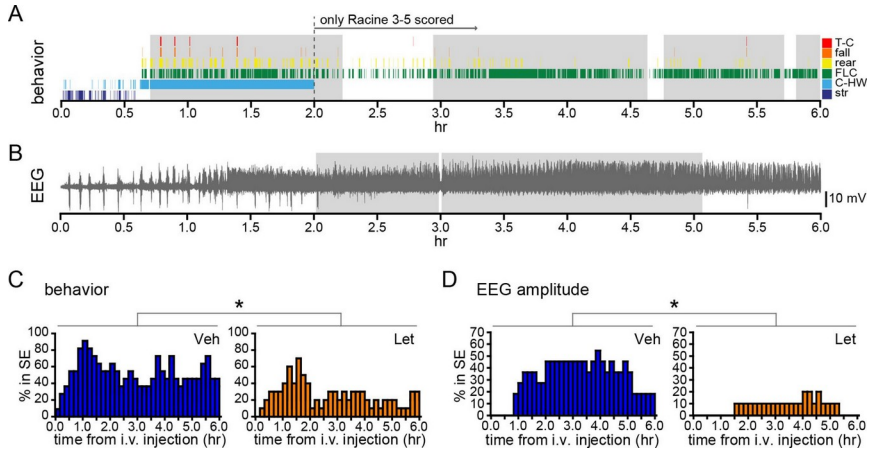
Roy et al. (2015), Fig. 1.

A prolonged seizure can sustain its own excitation

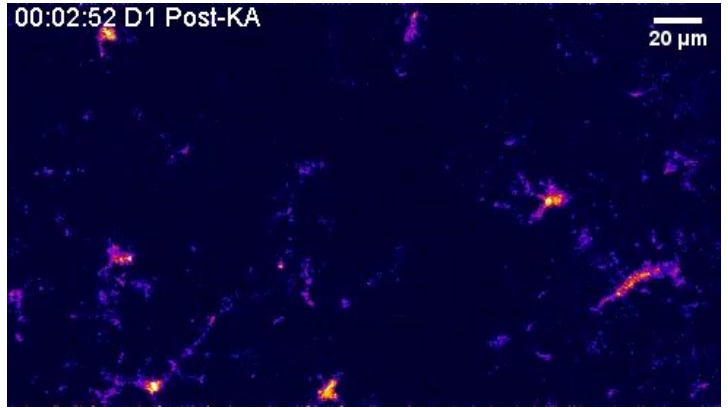
Status epilepticus is a prolonged seizure or repeated seizures without adequate recovery between them. It is a medical emergency because sustained abnormal activity can disturb breathing, circulation, metabolism, and neuronal function. It requires rapid clinical assessment and treatment.

During prolonged excitation, neurons release transmitters and experience large changes in ion movement. Calcium entry can activate enzymes that damage cellular structures. Energy demand also increases while the tissue must restore the ion gradients required for normal signaling.

Microglia, the brain's resident immune cells, also change after prolonged seizures. Separate imaging one day after kainate-induced status epilepticus shows calcium signals spreading among these cells. The recording follows the cellular response after the episode rather than electrical firing during the seizure itself.



Sato and Woolley (2016), Fig. 8.



Microglial calcium after status epilepticus. Umpierre et al. (2020).

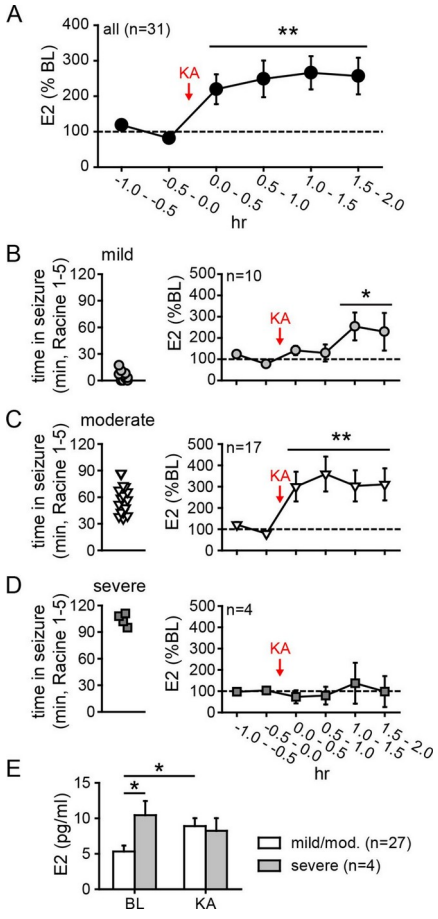
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Brain tissue can synthesize a seizure-modulating hormone

Estradiol is not produced only by peripheral reproductive organs. Neurons and other brain cells can synthesize steroid signals locally. Aromatase is the enzyme that converts suitable androgen precursors into estrogens, allowing local production to influence neural activity.

In a rat model of status epilepticus, researchers induced seizures with kainic acid. They measured hippocampal estradiol and recorded seizure activity. The hippocampus is relevant because its recurrent circuitry can participate in prolonged seizure activity.

Hippocampal estradiol increased during the experimentally induced seizures. Animals with more severe seizures had higher measured estradiol levels. This observation motivated an intervention targeting synthesis, rather than assuming that the hormonal change was only a consequence.



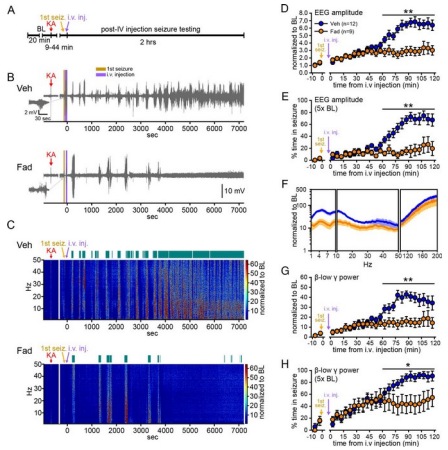
Sato and Woolley (2016), Fig. 4.

Aromatase inhibition reduces seizure escalation

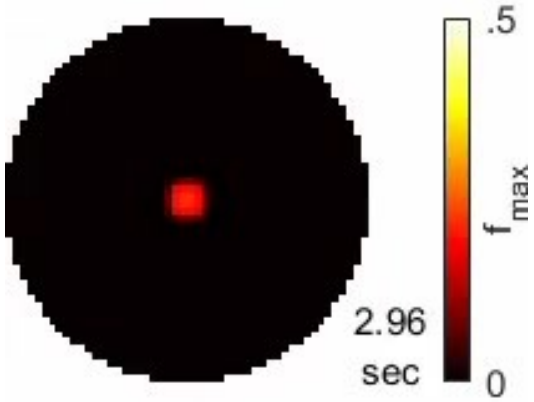
The investigators administered an aromatase inhibitor after seizure activity had begun. This timing tested an intervention during an established episode. The inhibitor reduced the enzyme's ability to generate estrogen from its precursor molecules.

Both electrographic and behavioral seizure measures decreased in the treated rats. Electrographic measures describe recorded electrical activity, whereas behavioral measures describe what the animal does. Agreement between the two measures strengthened the description of seizure suppression.

In a noise-free cortical model, a brief stimulus starts a symmetric expanding seizure. Rapid waves travel inward and meet at the center because the simulated tissue is symmetric. This model examines propagation geometry, whereas aromatase inhibition tests a biochemical mechanism in living animals.



Sato and Woolley (2016), Fig. 1.



Simulated expanding seizure front. Liou et al. (2020).

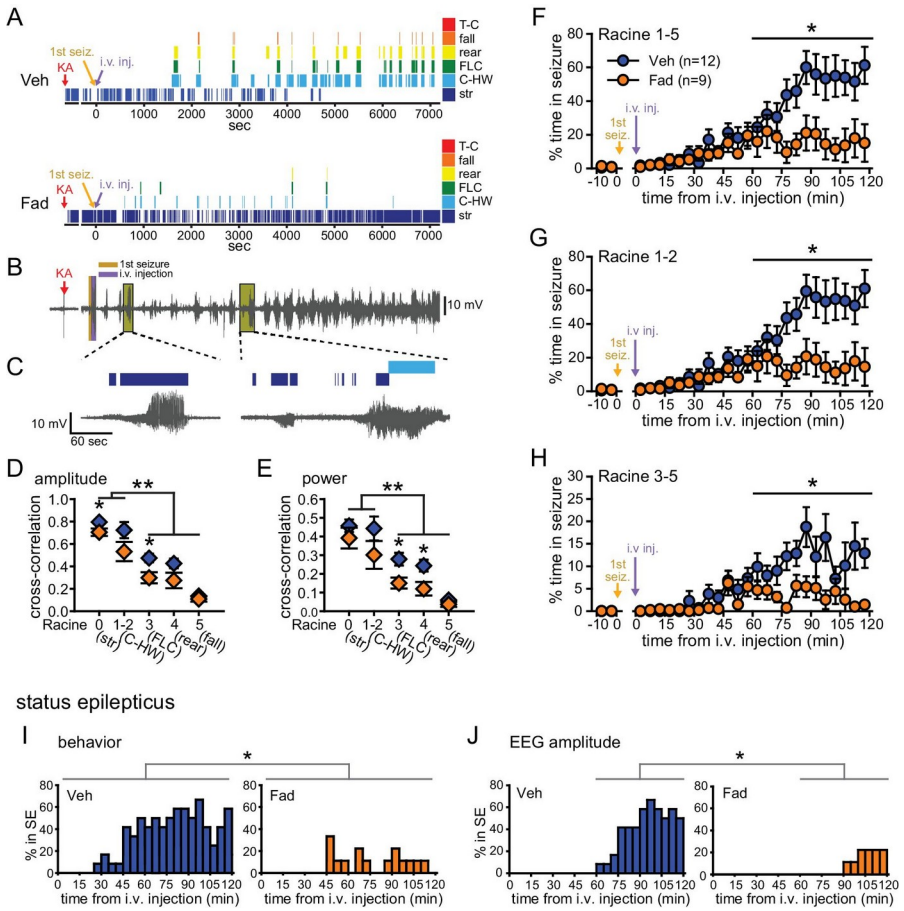
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A seizure aura reflects the network first recruited

An aura can be a subjective symptom at the beginning of a focal seizure. A change in smell, a visual experience, or an unusual internal sensation reflects the function of the tissue initially affected. It is part of the seizure itself. The first recruited network can process vision, smell, bodily sensation, or memory, so the experience provides an anatomical clue to the starting region.

Activity beginning in a motor region can produce movement in the corresponding body part. If recruitment spreads along the cortical representation, additional muscles may become involved. The sequence of symptoms can therefore provide information about the affected network.

A person's description remains important even when an episode is brief. An EEG obtained between events may not capture the same activity. Clinical interpretation combines the sequence of symptoms with recordings, imaging, and the circumstances surrounding the event.



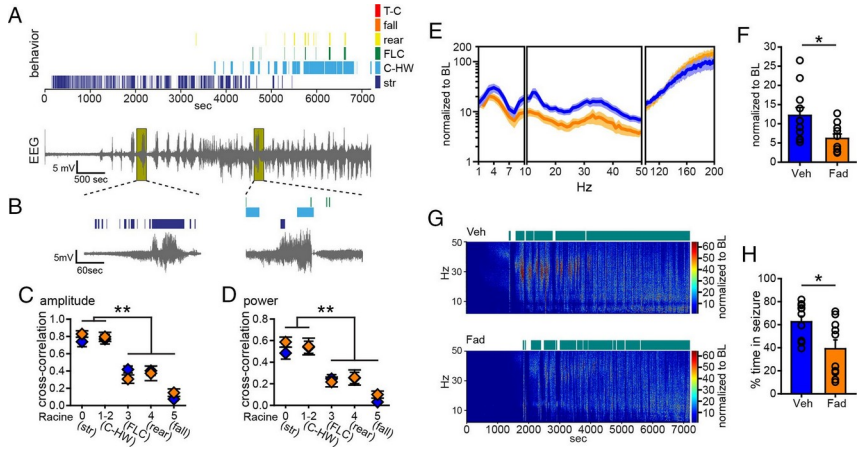
Sato and Woolley (2016), Fig. 2.

Local inhibition locates the relevant steroid synthesis

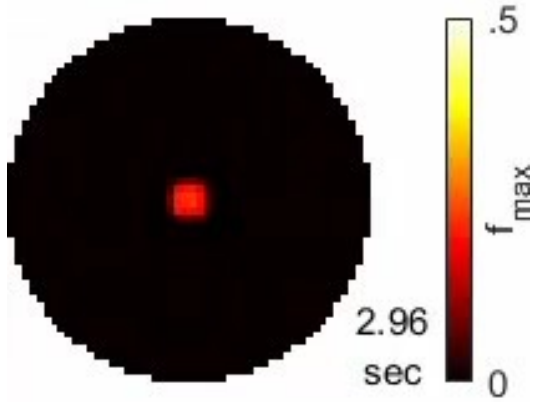
A systemic drug can reach many organs and brain regions. To test the hippocampus more directly, the researchers also delivered an aromatase inhibitor into hippocampal tissue. This narrowed the anatomical location of the manipulated synthesis pathway.

Local inhibition suppressed seizures in the rat model. The result made a peripheral endocrine explanation less sufficient, because the intervention acted within a brain region participating in the episode. The experiment directly changed local enzymatic activity.

Adding random input fluctuations changes how waves travel within the simulated seizure territory. Their meeting point drifts away from the center, and the later activity pattern becomes asymmetric. The changed input tests network dynamics rather than the local steroid-synthesis intervention used in the animal experiment.



Sato and Woolley (2016), Fig. 5.



Simulated uneven seizure propagation. Liou et al. (2020).

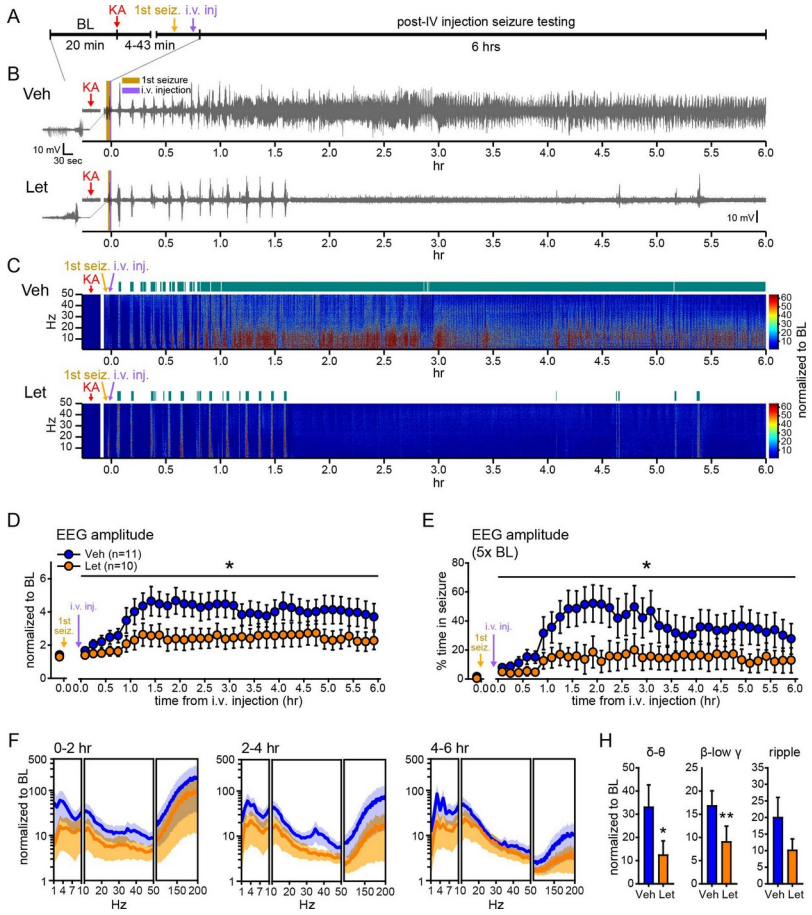
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Hormonal signaling changes on a rapid timescale

Steroid hormones can regulate gene expression, but they can also modify neuronal activity through faster signaling mechanisms. The seizure study examined acute estrogen synthesis during an ongoing event, rather than only long-term endocrine effects.

Kainic-acid seizures increased hippocampal estradiol during the episode. Reducing synthesis after onset reduced subsequent seizure expression. The relevant experimental sequence therefore included induction, local hormone production, and a later change after enzymatic inhibition.

The model used experimentally induced status epilepticus in rats. Its measurements establish the intervention's effect in that preparation. They do not make aromatase inhibitors a replacement for established emergency seizure treatment in people.



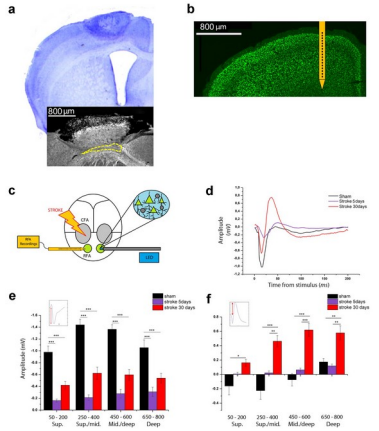
Sato and Woolley (2016), Fig. 7.

A stroke interrupts the supply needed for signaling

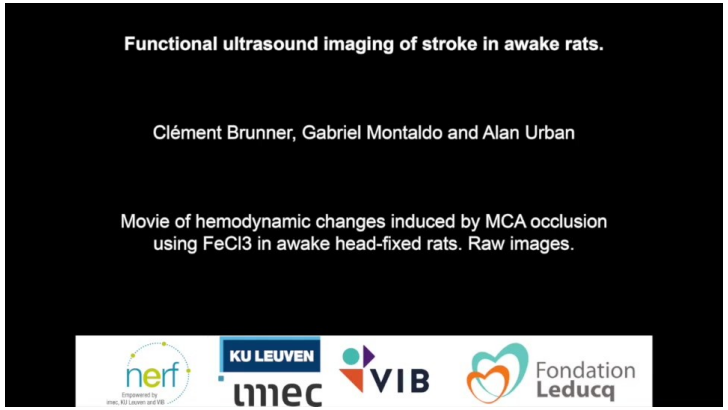
An ischemic stroke occurs when blood flow to part of the brain is obstructed. A hemorrhagic stroke involves bleeding from a damaged vessel. These mechanisms require different immediate treatments, even when both produce a sudden loss of neurological function.

Without adequate oxygen and glucose, neurons cannot sustain the ATP production needed to restore ion gradients. Membranes depolarize, extracellular glutamate can increase, and excessive receptor activation permits harmful calcium entry. This process is called excitotoxicity.

Ferric chloride can experimentally obstruct the middle cerebral artery in a rat. Ultrasound imaging follows the resulting change in regional blood-flow signals while the awake animal is head-fixed. The recording makes the vascular interruption visible before a later assessment of tissue damage or recovery.



Spalletti et al. (2017), Fig. 1.



Rat blood flow during arterial occlusion. Brunner et al. (2023).

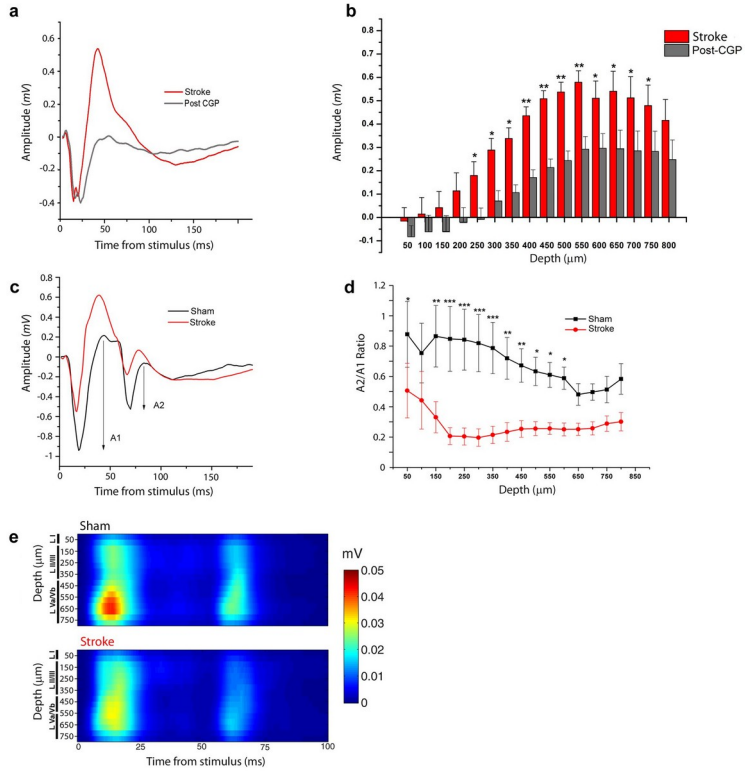
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A focal motor stroke alters activity in both hemispheres

Researchers produced a focal stroke in mouse forelimb motor cortex and measured changes in the remaining motor network. The damaged hemisphere retained surrounding tissue that could still participate in movement. The opposite hemisphere also continued to send signals across the corpus callosum.

The contralesional hemisphere is the side opposite the lesion. After the experimental stroke, it exerted stronger inhibitory influence over tissue near the lesion. This influence acted through communication between the hemispheres rather than direct destruction of the remaining tissue.

The investigators therefore combined a behavioral intervention with manipulation of this interhemispheric influence. The aim was to change the activity conditions under which the spared motor tissue practiced movements, while measuring the movements themselves.



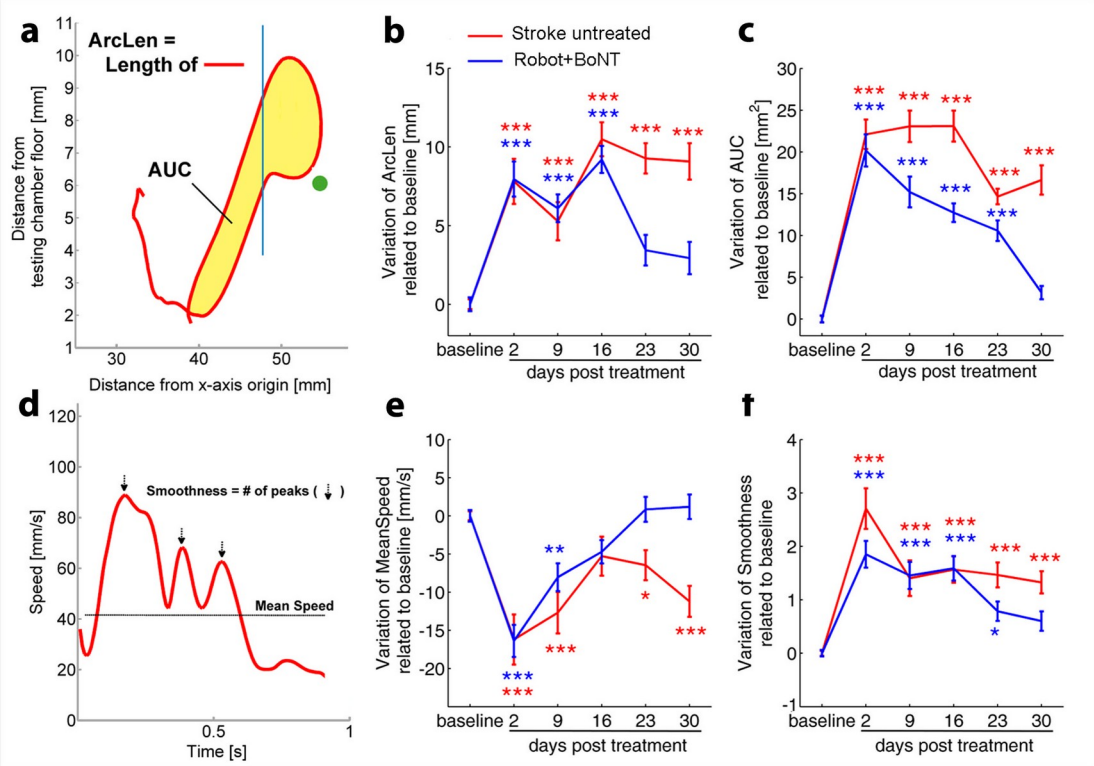
Spalletti et al. (2017), Fig. 3.

Robotic training makes repeated movement measurable

A robotic platform allowed mice to perform repeated forelimb movements under controlled conditions. The apparatus recorded features of the movement, making it possible to compare performance before injury and during recovery rather than relying only on a final success count.

A limb can reach a target by using a different movement strategy after injury. Kinematic analysis measures the trajectory and timing of the movement. It can therefore distinguish recovery toward the earlier pattern from compensation using an altered pattern.

The study combined these detailed measurements with broader motor tests. Restored task completion and restored movement organization were related outcomes, but the recorded trajectories supplied additional evidence about how the forelimb was being controlled.



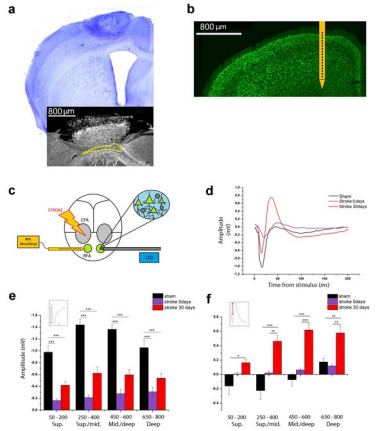
Spalletti et al. (2017), Fig. 8.

Reperfusion and rehabilitation act at different stages

Acute stroke care first identifies whether a vessel is blocked or bleeding. In selected ischemic strokes, medicines that dissolve clots or procedures that remove them can restore blood flow. Eligibility depends on imaging, timing, and clinical assessment.

Restoring blood flow aims to rescue threatened tissue. Rehabilitation later trains actions using tissue that survived the event. These interventions do not have the same biological target: one addresses vascular supply, while the other changes the use of remaining neural circuits.

Blood vessels also respond to low oxygen through their lining cells, called endothelial cells. In a separate vessel preparation, lowering oxygen increases brief calcium signals through TRPA1 channels. These local signals occur on a much shorter timescale than the movement practice used during rehabilitation.



Spalletti et al. (2017), Fig. 1.



Low-oxygen endothelial calcium signals. Pires et al. (2018).

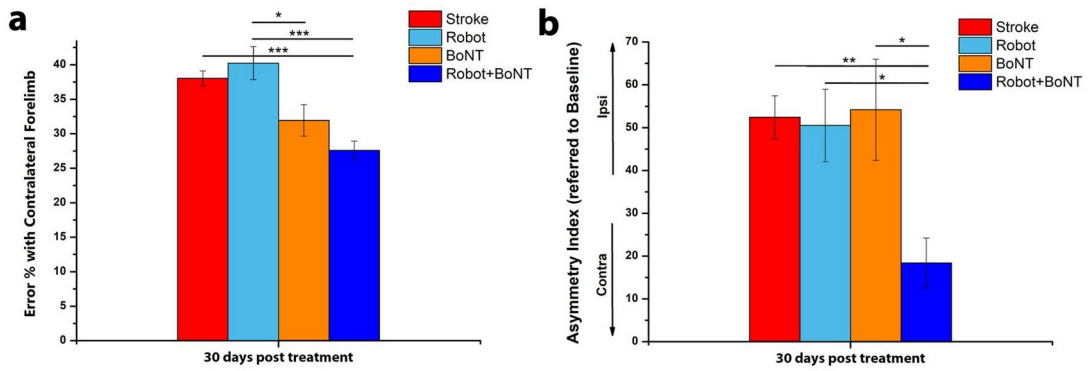
Play video

Training and temporary inactivation work together in mice

The researchers reversibly inactivated contralesional cortex while providing intensive robotic training. Reversible inactivation temporarily reduces activity, allowing an experimental test of the opposite hemisphere's contribution without permanently removing that tissue.

The combined protocol improved motor performance and manual dexterity. Recorded movement patterns moved toward the pre-stroke patterns. This outcome required assessing how the animals moved, rather than treating every successful reach as equivalent.

Recovery was accompanied by reduced transcallosal inhibition and changes in molecular constraints on plasticity near the lesion. The experiment manipulated the activity environment and provided structured practice, so both network state and repeated experience were part of the treatment.



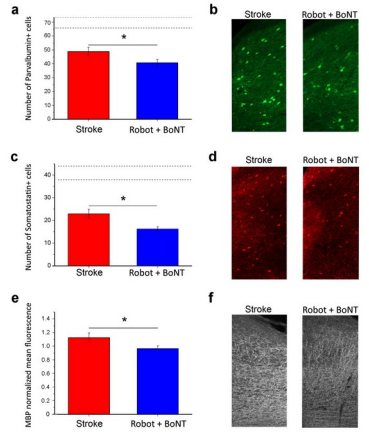
Spalletti et al. (2017), Fig. 7.

Recovery recruits spared tissue around the lesion

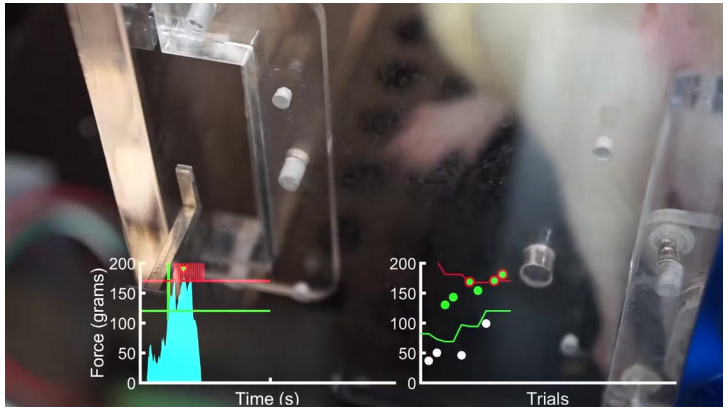
The combined mouse intervention did not regenerate the destroyed cortical region. Instead, it changed how surviving motor circuitry participated in trained behavior. Tissue near the lesion received a different balance of interhemispheric influence while the animals practiced.

A temporary reduction in contralesional activity could create conditions favorable for using the affected side's surviving circuits. The repeated exercises then supplied the movement experience needed to modify those circuits. The two components operated during the same recovery period.

A separate rehabilitation study paired vagus-nerve stimulation with strong forelimb pulls after rat spinal-cord injury. After six weeks, the rat pulls a handle while traces track its force and the stimulation threshold. This preparation tests recovery after a spinal lesion rather than the cortical stroke shown in the figure.



Spalletti et al. (2017), Fig. 9.



Forelimb rehabilitation with timed nerve stimulation. Ganzer et al. (2018).

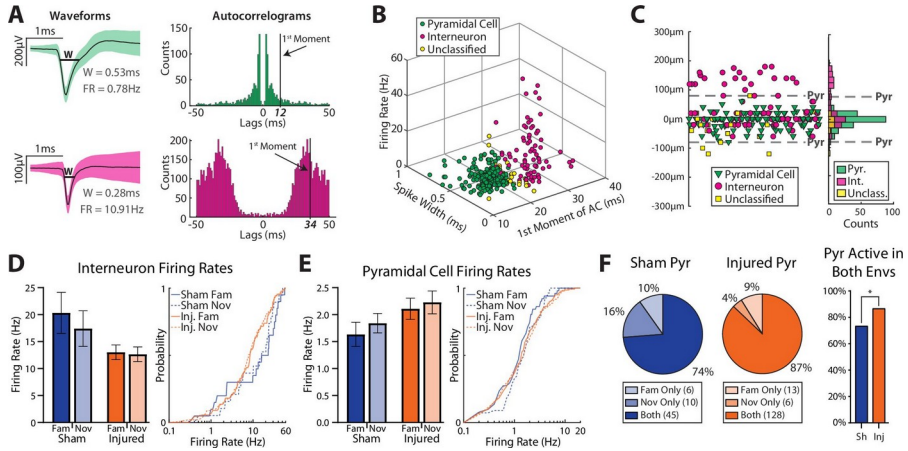
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Mechanical injury can disconnect apparently intact tissue

Traumatic brain injury results from an external mechanical force. A penetrating injury directly disrupts tissue, while a closed-head injury can damage the brain without opening the skull. Acceleration and rotation can stretch axons and injure blood vessels.

The site of impact is called the coup, and injury caused by the brain contacting the opposite side is the contrecoup. Distributed axonal injury can also occur away from either location. Damage therefore need not be confined to a single visible bruise.

Object use provides one clinical measure of function after severe brain injury. In the masked recording, a participant holds a comb and spoon but cannot demonstrate their use before zolpidem. Holding an object and using it appropriately require different combinations of movement and purposeful behavior.



Adam et al. (2026), Fig. 3.



Masked object-use assessment before zolpidem. Williams et al. (2013).

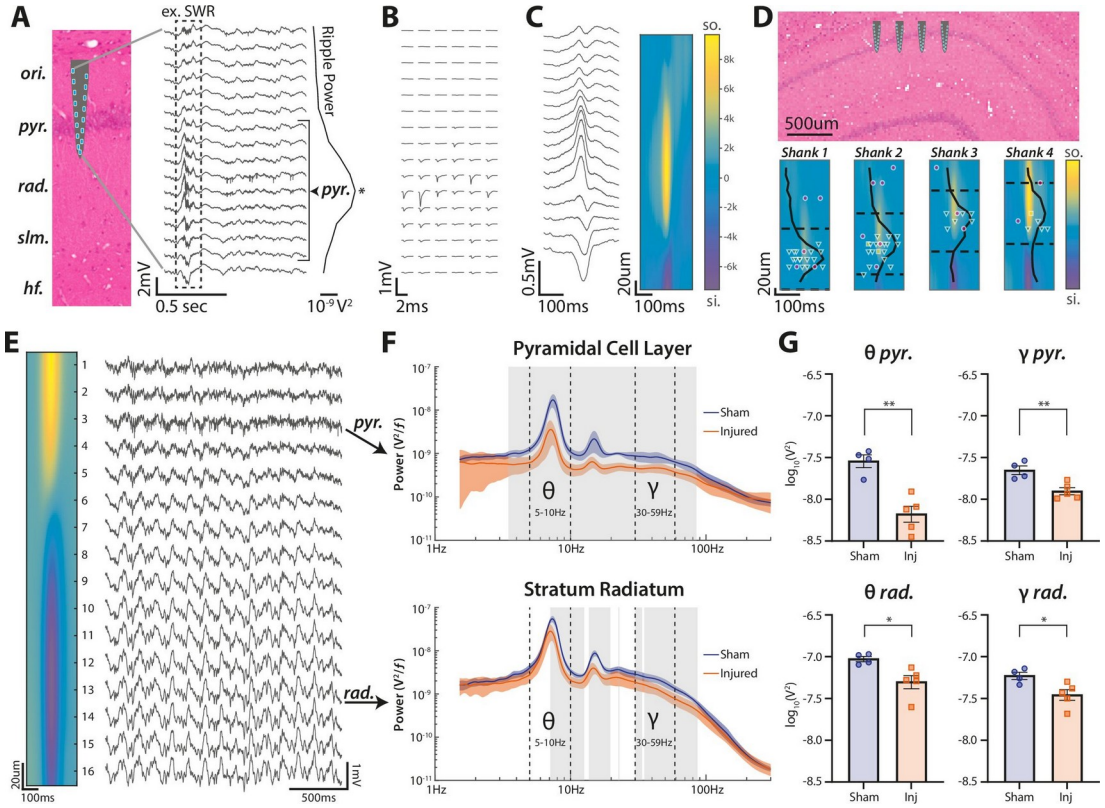
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Experimental injury disrupts hippocampal timing

Researchers recorded hippocampal activity in rats after experimental traumatic brain injury. The hippocampus contributes to forming and retrieving memories. Its neurons normally coordinate their activity with oscillations that organize signaling over several different timescales.

High-density electrodes sampled activity across layers of area CA1. This arrangement distinguished the signals recorded in different anatomical compartments. Measurements included local field potentials and the timing of identified neuronal spikes relative to ongoing rhythms.

The injured animals showed reduced oscillatory power and layer-specific disruption of theta–gamma coupling. Coupling describes how the amplitude of a faster rhythm varies with the phase of a slower rhythm, rather than simply whether both rhythms are present.



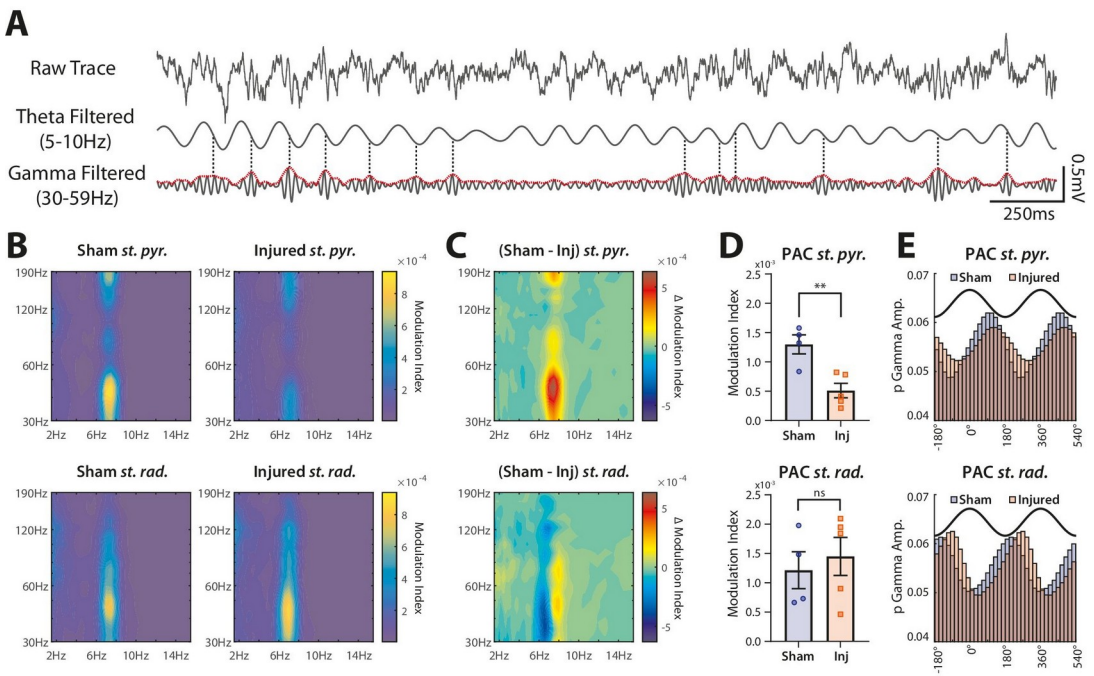
Adam et al. (2026), Fig. 1.

Theta phase helps organize faster bursts of activity

Theta is a relatively slow hippocampal rhythm, while gamma describes faster oscillations. A phase identifies a position within an oscillation's cycle. When gamma amplitude changes systematically with theta phase, faster activity is organized within the slower cycle.

In the injured rats, this phase–amplitude organization was weakened in particular CA1 layers. A smaller coupling measure does not require complete absence of gamma activity. It means that its timing relative to theta became less consistently organized.

Layer-specific recordings were important because different inputs reach different parts of hippocampal neurons. A recording averaged across the structure could obscure the compartment in which temporal coordination changed most strongly after injury.



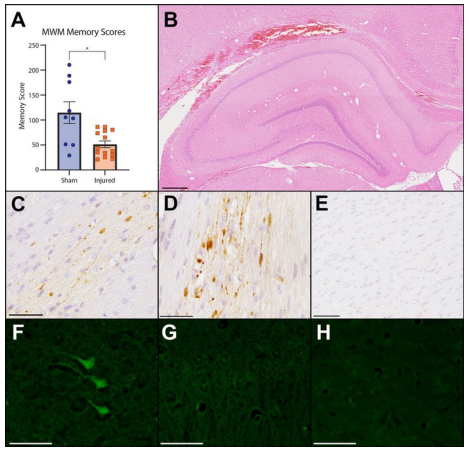
Adam et al. (2026), Fig. 2.

Axonal damage and circuit timing affect different scales

An axon carries action potentials from one region to another. Stretching or damaging axonal structures can interrupt transport and signal propagation. A connected region may therefore receive abnormal input even when its own neurons were not directly struck.

Memory also requires coordination among surviving neurons. The timing of excitation and inhibition determines when a cell is likely to fire and which inputs coincide. Physiological dysfunction can consequently persist in tissue that remains anatomically present.

The same participant is tested again after zolpidem and demonstrates use of the comb and spoon. The assessment records a change in purposeful object use and posture. This individual drug response supplies a behavioral measurement rather than an image of repaired axons or a general treatment result.



Adam et al. (2026), Fig. 7.



Masked object-use assessment after zolpidem. Williams et al. (2013).

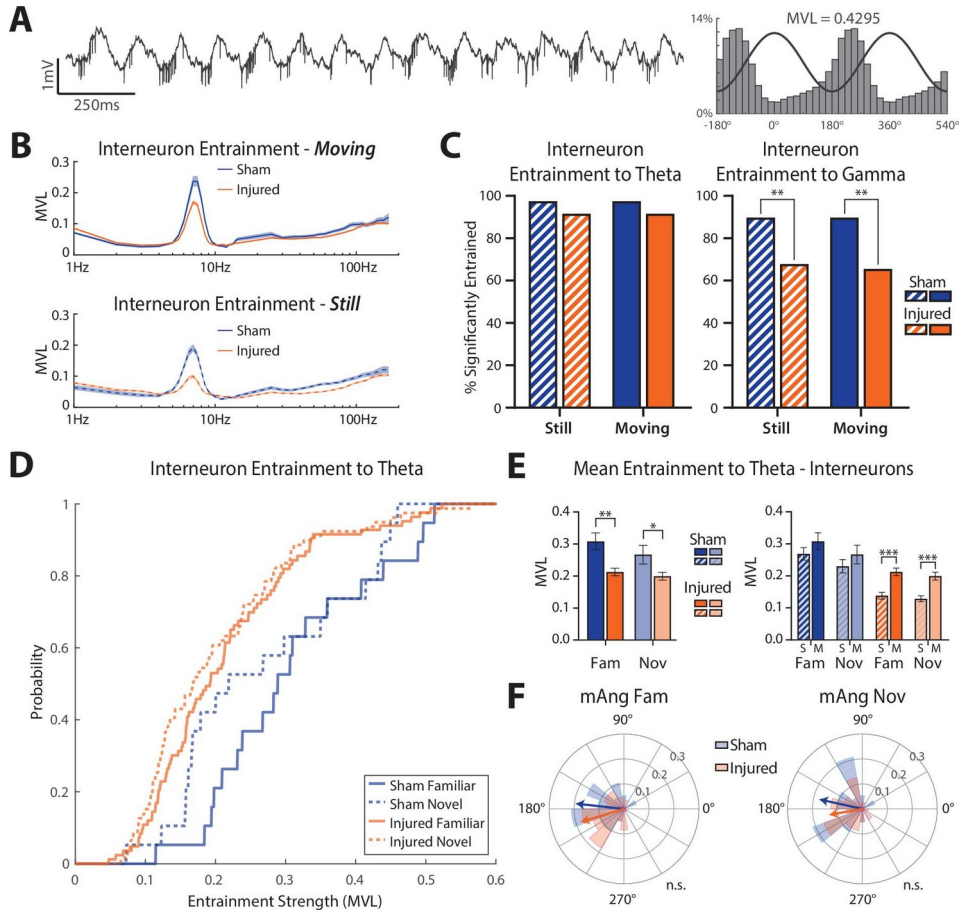
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Interneurons lose precise entrainment after injury

Interneurons help control when other hippocampal neurons fire. In the injured rats, interneuron spikes were less strongly entrained to theta and gamma rhythms. Entrainment means that spikes occur preferentially at particular phases of a recurring signal.

The researchers also examined periods with stronger theta power. During those periods, both interneurons and pyramidal neurons showed stronger theta entrainment than during weaker-theta periods. Injury altered the activity pattern without eliminating all state-dependent organization.

These analyses separated neuronal type, oscillatory frequency, and brain state. The result was a specific timing abnormality in hippocampal circuitry, rather than a claim that every neuron became uniformly more active or uniformly less active.



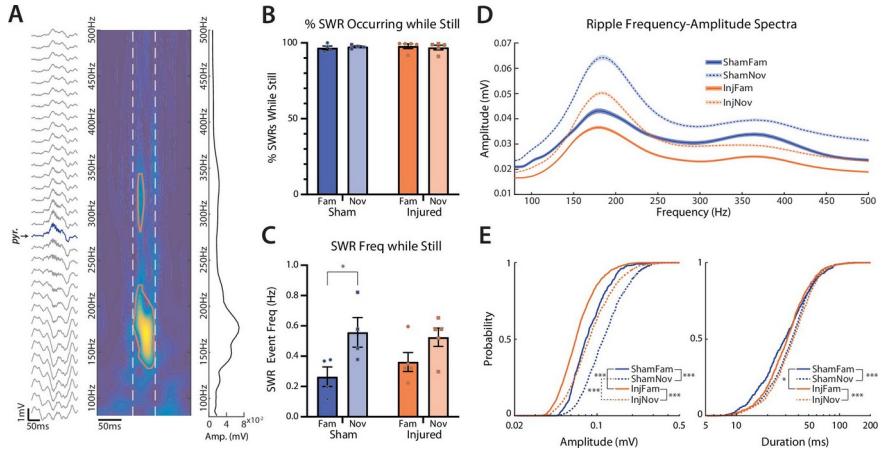
Adam et al. (2026), Fig. 4.

Quiet wakefulness reveals a second timing abnormality

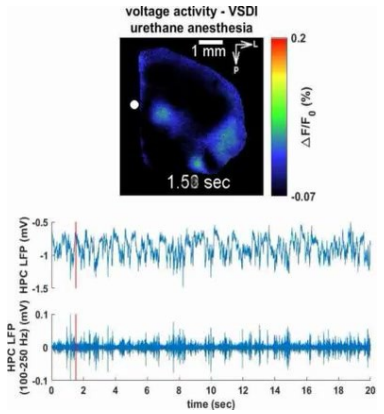
During quiet immobility, hippocampal networks generate brief events called sharp-wave ripples. These coordinated population events differ from the theta activity often prominent during exploration. The study examined both activity states after traumatic brain injury.

Ripple amplitudes were smaller in injured animals than in sham-treated animals. A sham group undergoes the comparison procedure without the experimental injury, helping distinguish effects of the lesion from effects of handling and surgery.

A separate recording pairs cortical voltage imaging with hippocampal electrical signals under urethane anesthesia. The lower filtered trace makes rapid ripple events visible beside slower cortical changes. This preparation demonstrates simultaneous measurements across regions, rather than the injured animals' quiet-waking condition.



Adam et al. (2026), Fig. 6.



Cortical voltage and hippocampal ripple signals. Abadchi et al. (2020).

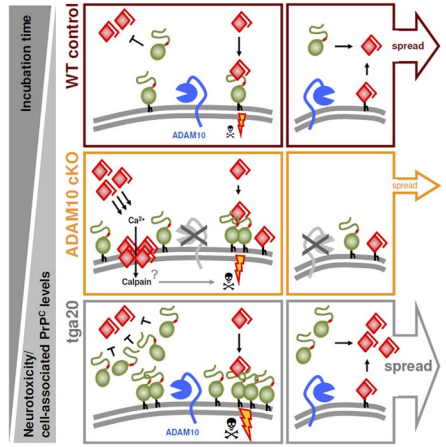
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Prion disease spreads a protein conformation

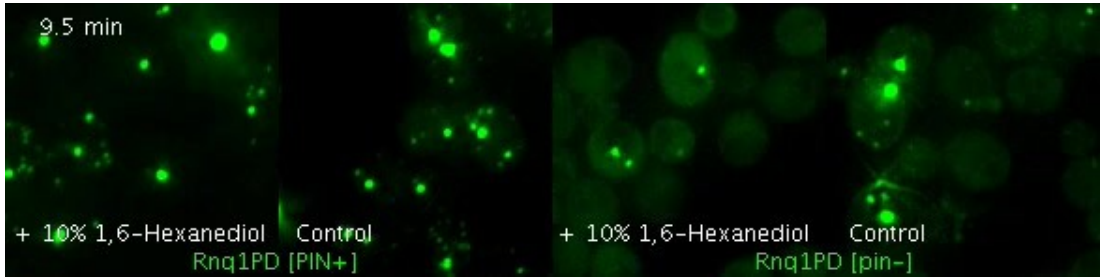
A prion is an infectious protein conformation that promotes conversion of normally folded protein into an abnormal form. The propagated feature is the protein's shape. This mechanism differs from replication of a bacterium or a virus carrying its own genome.

Prion diseases can be sporadic, inherited through variants in the PRNP gene, or acquired through exposure under particular conditions. The normal prion protein is made by the host's cells. Abnormal forms accumulate as normal molecules adopt the altered conformation.

Yeast provides another preparation for studying stable protein assemblies. Fluorescent imaging compares amyloid and non-amyloid forms of the Rnq1 protein after hexanediol exposure. Non-amyloid assemblies dissolve more readily; the yeast protein is different from the PrP protein responsible for mammalian prion disease.



Altmeppen et al. (2015), Fig. 9.



Yeast amyloid and non-amyloid assemblies. Kroschwald et al. (2015).

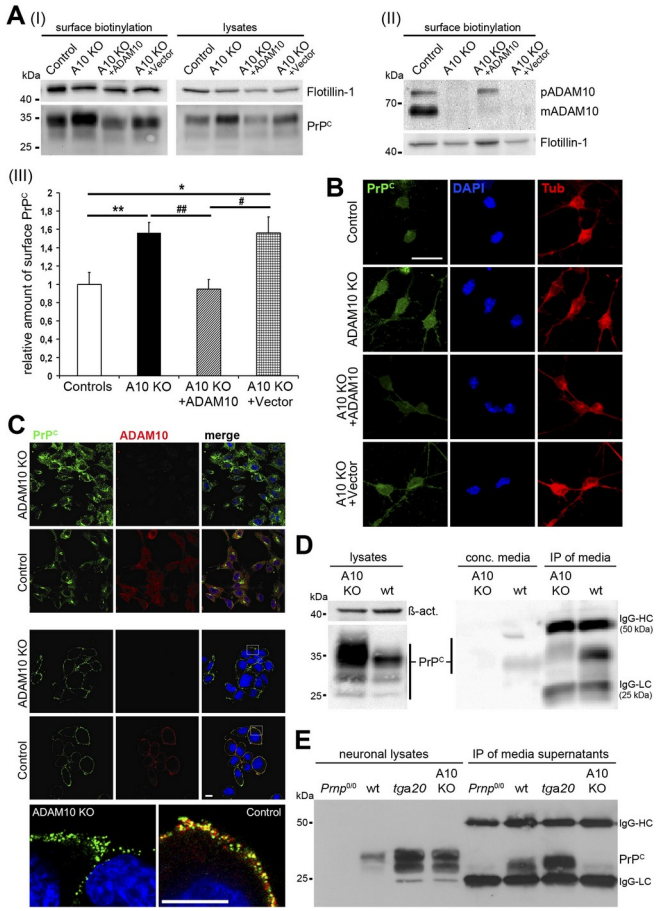
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ADAM10 can release prion protein from the cell surface

The normal prion protein is attached to the outside of the cell membrane. ADAM10 is an enzyme that can cut membrane-associated proteins, releasing an external portion. This process is called shedding because material leaves the cell surface.

Researchers changed ADAM10 activity in mouse models to examine its role in prion disease. Their experiments measured prion-protein processing and disease-related outcomes. The intervention altered handling of a host protein, rather than directly attacking a bacterial or viral genome.

Removing ADAM10 from forebrain neurons increased normal prion-protein levels after translation. After experimental prion infection, these mice developed disease sooner and accumulated more abnormal prion protein. The intervention therefore changed both the available host protein and the course of infection.



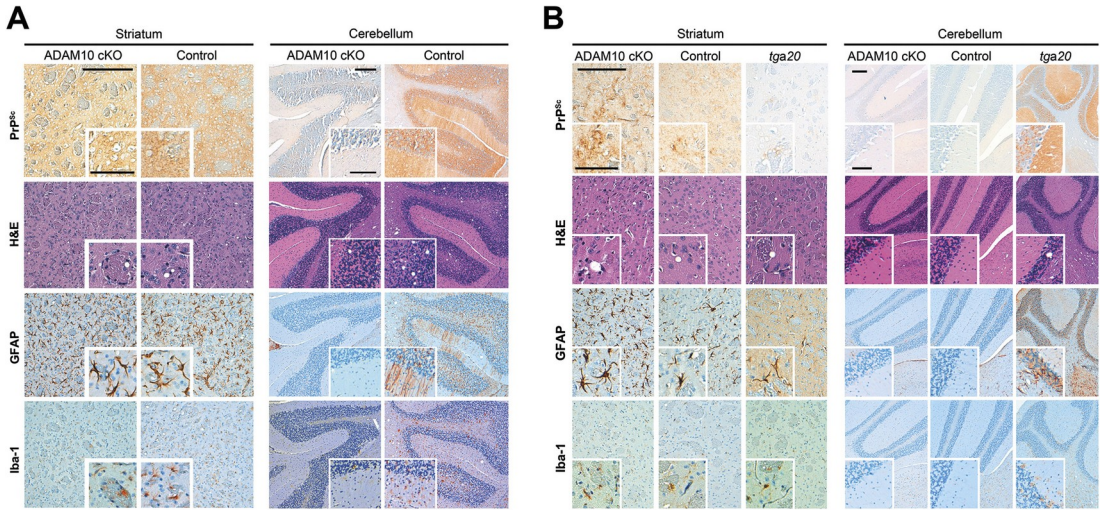
Altmeppen et al. (2015), Fig. 1.

Protein location changes access to conversion

Conversion requires contact between an abnormal template and susceptible host protein. A membrane attachment changes the physical location at which that contact can occur. Shedding changes the distribution of the same host protein without changing its gene sequence.

In the ADAM10 study, the investigators assessed the biochemical consequences of changing the sheddase. They also followed disease progression in animals, because altered protein processing in a sample does not by itself specify the eventual neurological outcome.

The absence of shedding also impaired the spatial spread of pathology, even while disease developed sooner. Local protein accumulation and propagation to other sites therefore changed in different directions. The experiment separated the concentration of susceptible protein from its distribution through tissue.



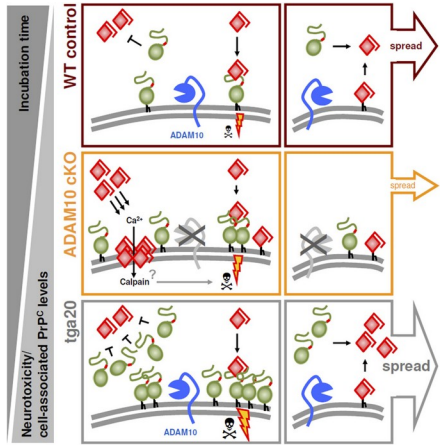
Altmeppen et al. (2015), Fig. 8.

Misfolded proteins are handled by several cellular systems

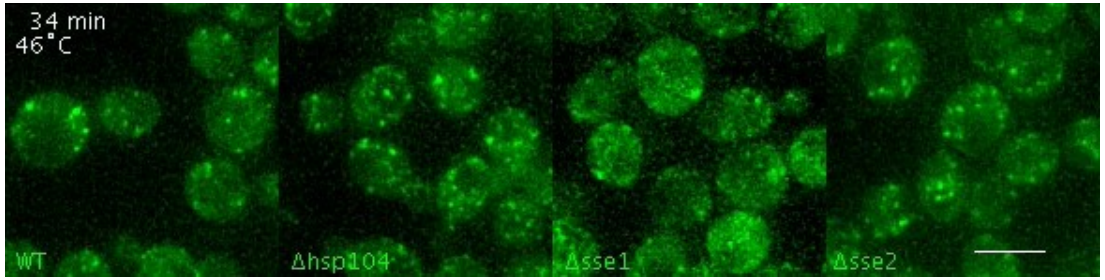
Cells contain machinery that folds proteins, transports them, and removes damaged material. A proteasome breaks down many tagged proteins, while lysosomes digest material delivered through membrane-bound pathways. These routes handle different substrates and cellular compartments.

Protein aggregation occurs when molecules associate into larger assemblies. An aggregate's size and shape can influence where it travels and which degradation pathway can access it. Visible deposits and smaller soluble assemblies can therefore have different biological effects.

Disaggregases are proteins that help take apart protein assemblies. In yeast recovering from heat shock, fluorescent stress granules dissolve differently when particular disaggregases are missing. The recording follows this cellular recovery process; these granules are different structures from infectious mammalian prions.



Altmeppen et al. (2015), Fig. 9.



Yeast stress-granule recovery after heat shock. Kroschwald et al. (2015).

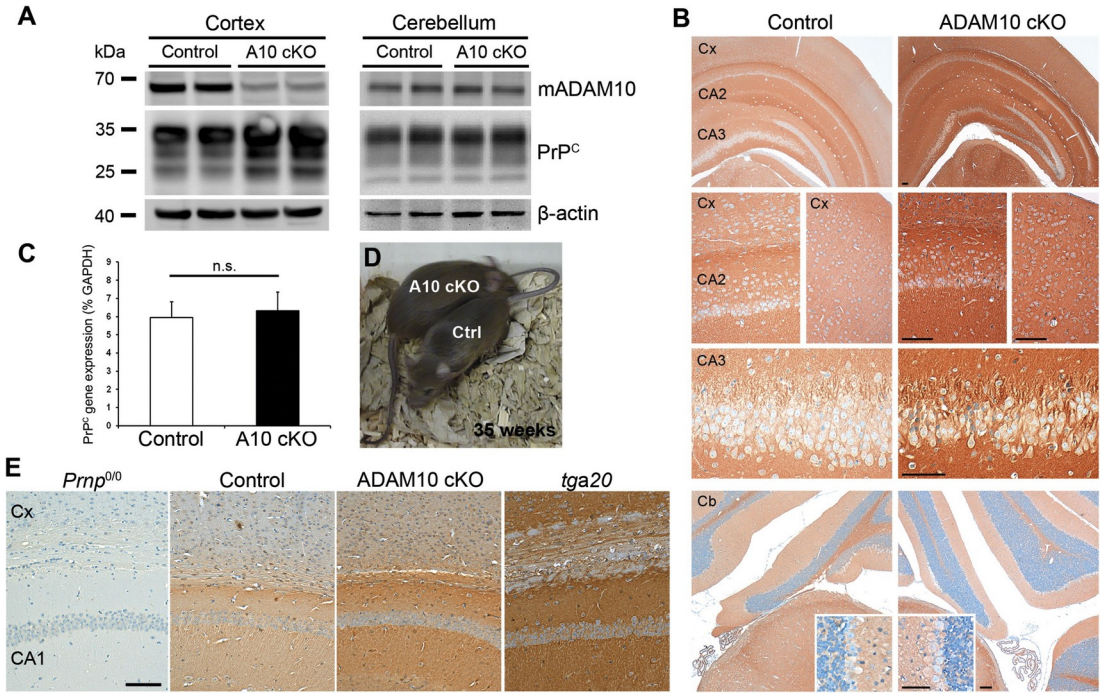
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Shedding changes where prion protein accumulates

A tissue can contain a similar total amount of prion protein while distributing it differently between surface and released forms. The ADAM10 experiments therefore examined protein processing rather than inferring surface availability from a single total-protein measurement.

The relevant enzyme acts at a specific processing step. Altering that step changes the forms present around cells and can change the progression of experimental prion disease. Measurements of released protein identify a consequence that a gene-expression assay alone would miss.

The increased normal prion-protein level followed a post-translational change, meaning that it arose after the protein had been made. ADAM10 removal altered processing at the cell surface. The researchers therefore measured protein forms and localization alongside RNA expression and disease progression.



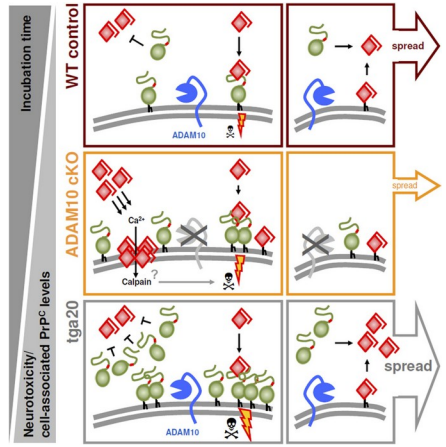
Altmeppen et al. (2015), Fig. 2.

Prion interventions target a host-dependent process

Host cells supply the normal protein that prions convert. Reducing its availability, altering its processing, or interfering with conversion are therefore distinct experimental strategies. Each strategy targets a different step between protein production and tissue damage.

An inherited PRNP variant changes the protein sequence, whereas a sporadic case does not require a recognized inherited variant. Acquired transmission introduces an abnormal conformational template. These routes can converge on protein misfolding while beginning with different events.

At the same time after prion inoculation, an ADAM10-deficient mouse shows little activity and lacks a nest, while its littermate remains more active beside a formed nest. These behavioral observations accompany the molecular measurements. They describe progression in the experimental mice rather than a successful clinical intervention.



Altmeppen et al. (2015), Fig. 9.



Nesting and activity after prion inoculation. Altmeppen et al. (2015).

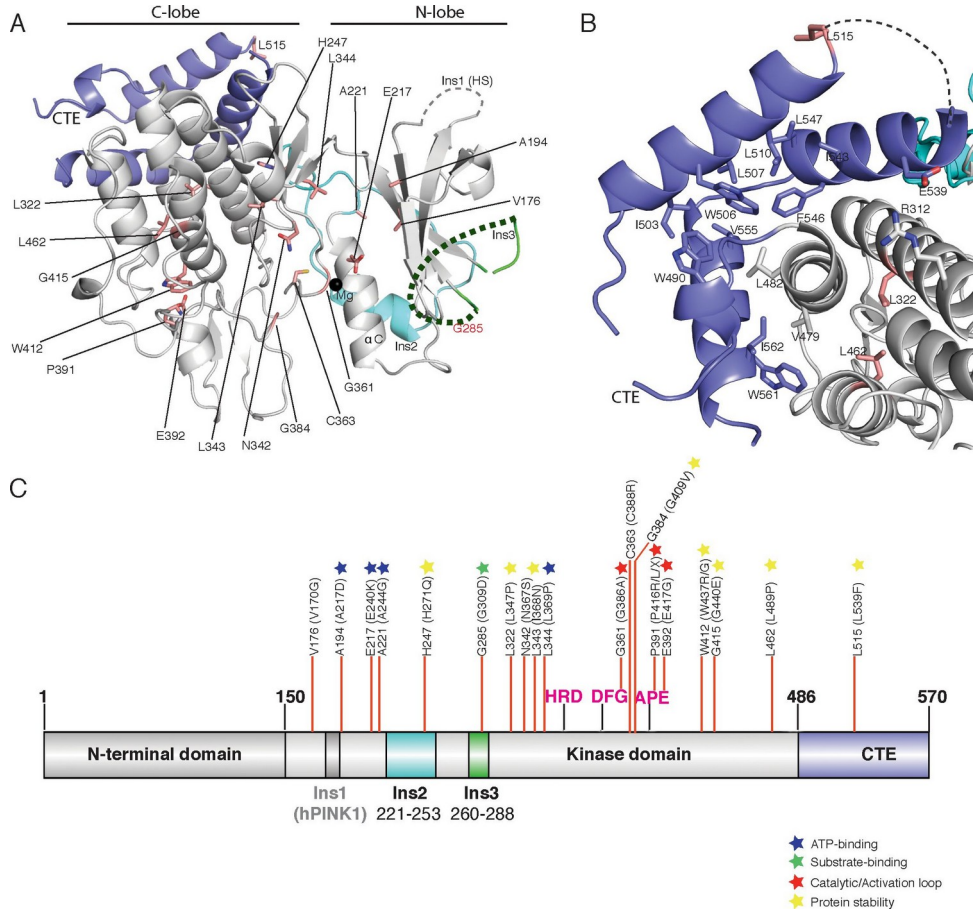
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Parkinsonian movement follows loss of nigrostriatal input

Parkinson's disease commonly produces slowed movement, rigidity, and resting tremor, alongside nonmotor symptoms. Dopaminergic neurons in the substantia nigra normally send axons to the striatum. Loss of this input changes signaling through basal-ganglia circuits involved in action.

Dopamine has different effects on striatal neurons carrying different receptor types. Reducing dopamine does not simply switch off every striatal cell. It changes the balance of activity through pathways that facilitate, constrain, and organize movement.

Levodopa supplies a precursor that remaining neurons can convert into dopamine. It can improve symptoms without replacing the lost neurons. The distinction between symptomatic improvement and slowing the underlying degenerative process remains important when comparing treatments.



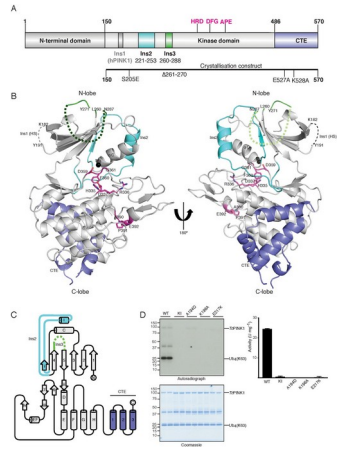
Kumar et al. (2017), Fig. 5.

PINK1 participates in mitochondrial quality control

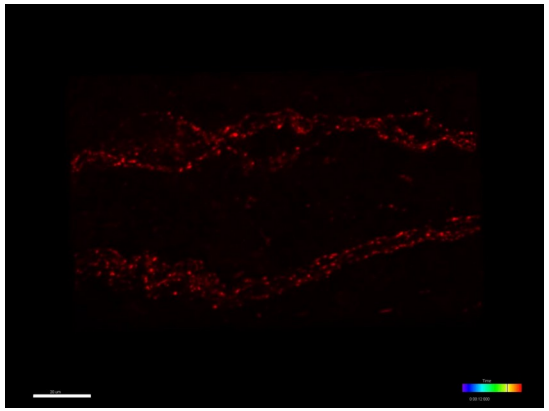
PINK1 is a protein kinase, an enzyme that transfers phosphate groups to other molecules. It participates in pathways responding to damaged mitochondria. Mitochondria generate much of a neuron's ATP, so their maintenance is especially important in energy-demanding cells.

Inherited loss-of-function variants in PINK1 can cause an early-onset form of Parkinson's disease. Researchers examined the structure of a beetle PINK1 protein to understand how this unusual kinase binds its targets and how disease-associated changes could impair its activity.

Mitochondria can be examined in living human intestinal nerve tissue as well as in laboratory cells. A voltage-sensitive fluorescent dye labels mitochondria in a control enteric ganglion. The changing signal comes from gut neurons, not the dopamine neurons that degenerate in the midbrain.



Kumar et al. (2017), Fig. 1.



Mitochondria in a control enteric ganglion. Desmet et al. (2017).

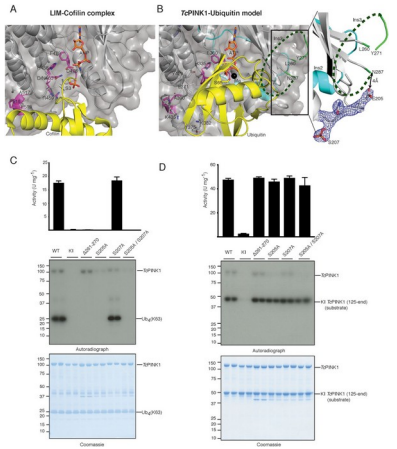
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PINK1 recognizes ubiquitin through a specialized surface

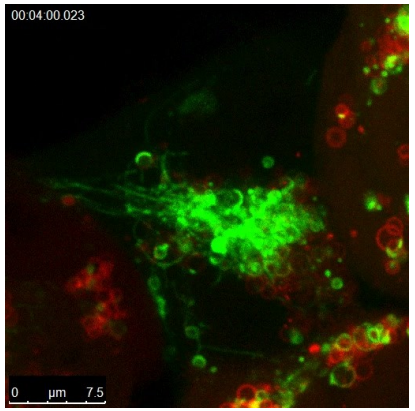
Ubiquitin is a small protein that can be attached to other proteins as a cellular signal. PINK1 phosphorylates ubiquitin and participates in activation of Parkin, an enzyme that attaches ubiquitin to suitable targets in mitochondrial quality-control pathways.

The structural study identified insertions that distinguish PINK1 from many other kinases. One insertion and an autophosphorylation event helped create a binding surface for ubiquitin. Autophosphorylation means that the kinase adds phosphate to itself.

Parkin participates in directing damaged mitochondria toward cellular disposal. Separate imaging follows fluorescent Parkin and the membrane-associated protein LC3 in cells lacking TBC1D15 after mitochondrial depolarization. Tubular LC3 structures move around the labeled organelles, revealing a later cellular stage than PINK1 substrate recognition.



Kumar et al. (2017), Fig. 2.



Parkin and LC3 after mitochondrial depolarization. Yamano et al. (2014).

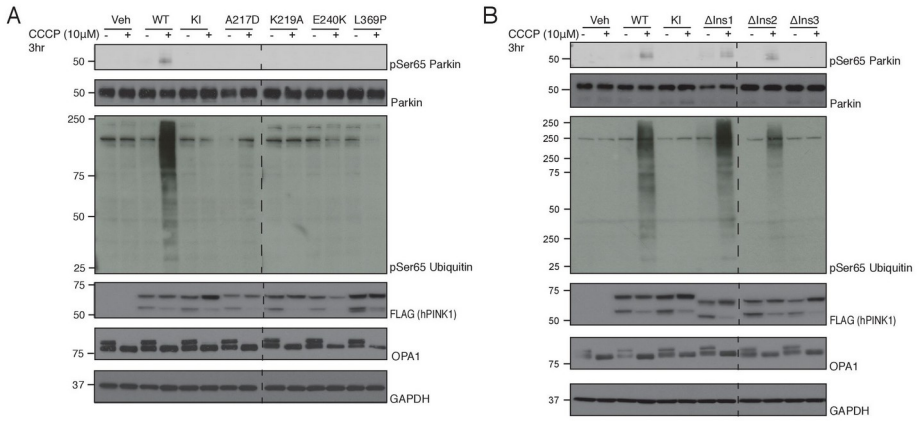
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Mitochondrial repair and dopamine act differently

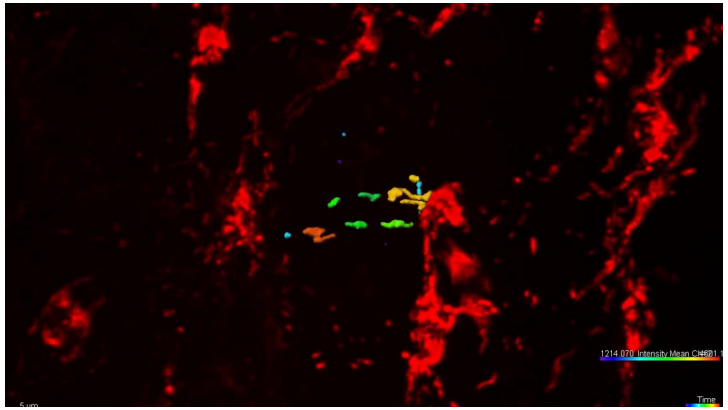
A mitochondrion supplies ATP and participates in calcium handling and cellular stress responses. Damaged mitochondria can generate abnormal reactive molecules and fail to meet local energy demand. Neurons require quality-control pathways to remove or repair compromised components.

PINK1 and Parkin participate in signaling that identifies damaged mitochondria for removal. This process is part of mitophagy, the selective disposal of mitochondria. It addresses organelle quality rather than directly increasing dopamine concentrations at striatal synapses.

The corresponding patient recording follows labeled mitochondria in an intestinal ganglion from a person with Parkinson's disease. Fluorescence varies over time, and a three-dimensional mask outlines the measured mitochondrial volume. This is an enteric-tissue measurement rather than a demonstration of restored movement after dopamine treatment.



Kumar et al. (2017), Fig. 4.



Mitochondria in a Parkinson enteric ganglion. Desmet et al. (2017).

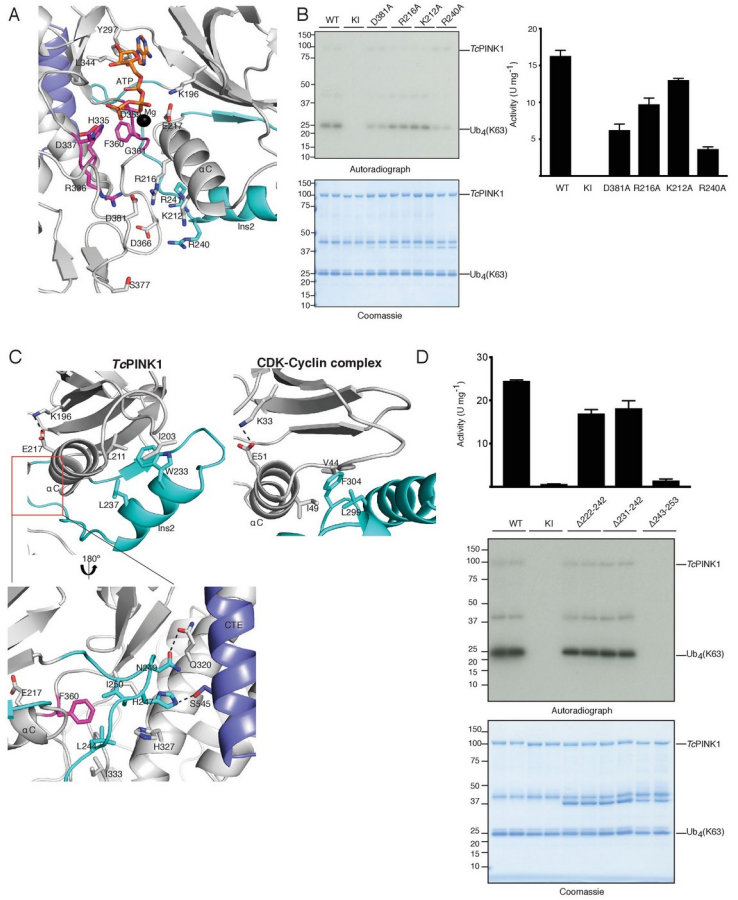
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Disease variants disrupt kinase architecture

A protein's activity depends on contacts among parts of its folded structure. The PINK1 researchers identified structural elements held together by interactions between an insertion and a distant loop. Disturbing these contacts reduced enzymatic activity in their experiments.

The structure provided an explanation for several disease-associated variants within the kinase domain. A variant may impair the protein's architecture or its ability to interact with substrates, rather than acting solely by reducing the amount of protein present.

Such experiments examine a specific biochemical defect. They connect a changed amino acid to altered kinase function through structural and activity measurements. The molecular result remains separate from the many additional processes operating in an intact human dopaminergic neuron.



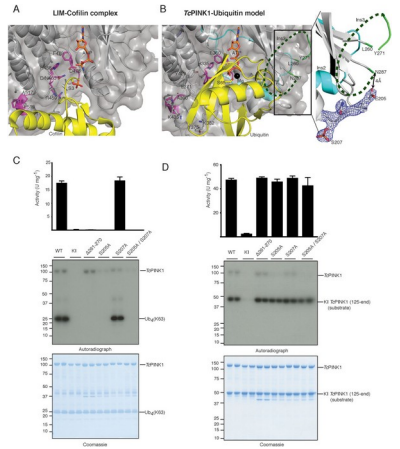
Kumar et al. (2017), Fig. 3.

Phosphorylation changes the signal carried by ubiquitin

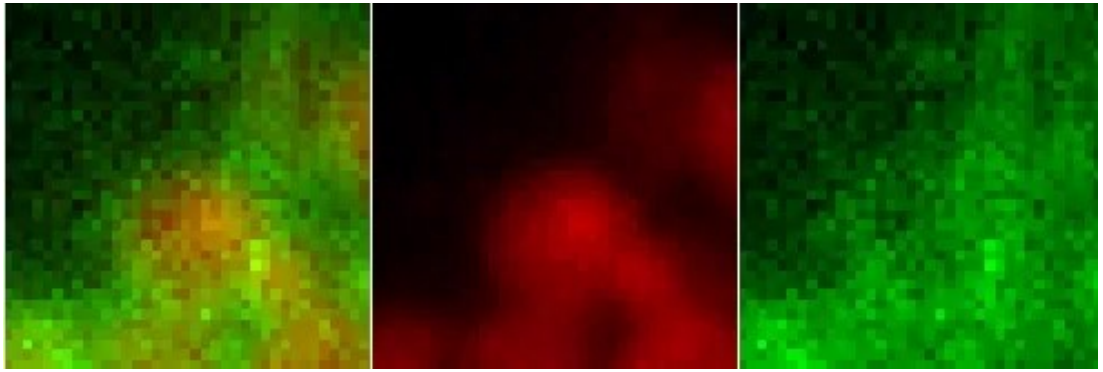
Adding a phosphate group changes a molecule's chemical properties and its interactions with other proteins. In the PINK1 pathway, phosphorylation of ubiquitin helps regulate Parkin activity. The modification changes a signaling state rather than synthesizing a new organelle.

The structural analysis showed how PINK1 can accommodate ubiquitin as a substrate. The investigators combined the structure with biochemical tests, allowing them to evaluate whether particular structural elements were required for the measured enzymatic activity.

Mitochondrial stress can also change the location of trafficking proteins. In a separate cell preparation, laser-induced stress draws fluorescent Rab5 toward labeled mitochondria. The recording follows Rab5 redistribution, a different event from the PINK1-dependent phosphate transfer measured in the structural experiments.



Kumar et al. (2017), Fig. 2.



Rab5 redistribution during mitochondrial stress. Hsu et al. (2018).

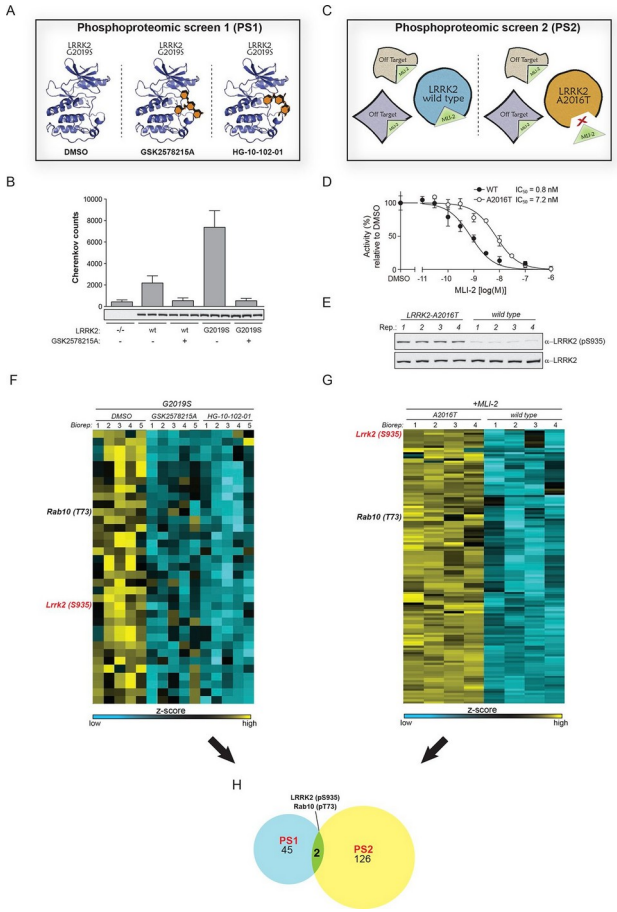
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LRRK2 changes trafficking proteins by phosphorylation

LRRK2 is another kinase implicated in inherited Parkinson's disease. Certain pathogenic variants increase its activity. Researchers searched systematically for proteins whose phosphorylation changed when LRRK2 activity was altered by genetic or pharmacological interventions.

They identified a subset of Rab proteins as direct LRRK2 substrates. Rab proteins are molecular switches involved in membrane trafficking, the movement and organization of membrane-bound compartments. Phosphorylation therefore affected proteins responsible for directing cellular cargo.

The study combined broad protein measurements with targeted tests. This distinguished candidate associations from direct enzymatic modification and identified a specific group of trafficking regulators rather than attributing every altered cellular pathway to LRRK2.



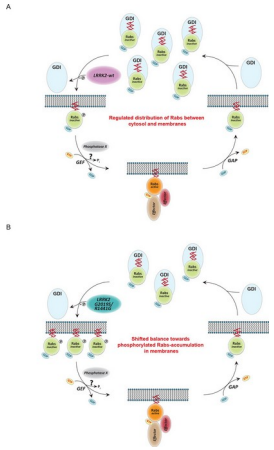
Steger et al. (2016), Fig. 1.

Rab switches help route membrane-bound cargo

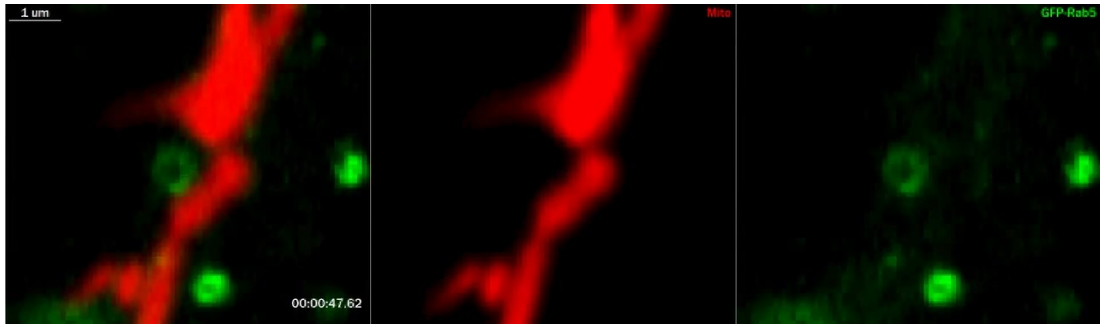
Cells transport many substances in vesicles, small membrane-bound compartments. Rab proteins help identify compartments and recruit partners needed for movement, attachment, and fusion. A Rab's interactions change as it switches between nucleotide-bound states.

A kinase can modify this trafficking system by adding phosphate to a Rab protein. The added group can change binding to regulatory proteins. The consequence is a change in how the Rab is handled, rather than a direct physical movement of the vesicle by the kinase.

Rab5 labels early endosomes, membrane compartments that sort material taken into a cell. Live imaging shows these compartments moving near mitochondria labeled in another color. Their positions change over time, allowing membrane trafficking to be examined as a physical movement rather than only a protein measurement.



Steger et al. (2016), Fig. 7.



Endosome movement beside mitochondria. Hsu et al. (2018).

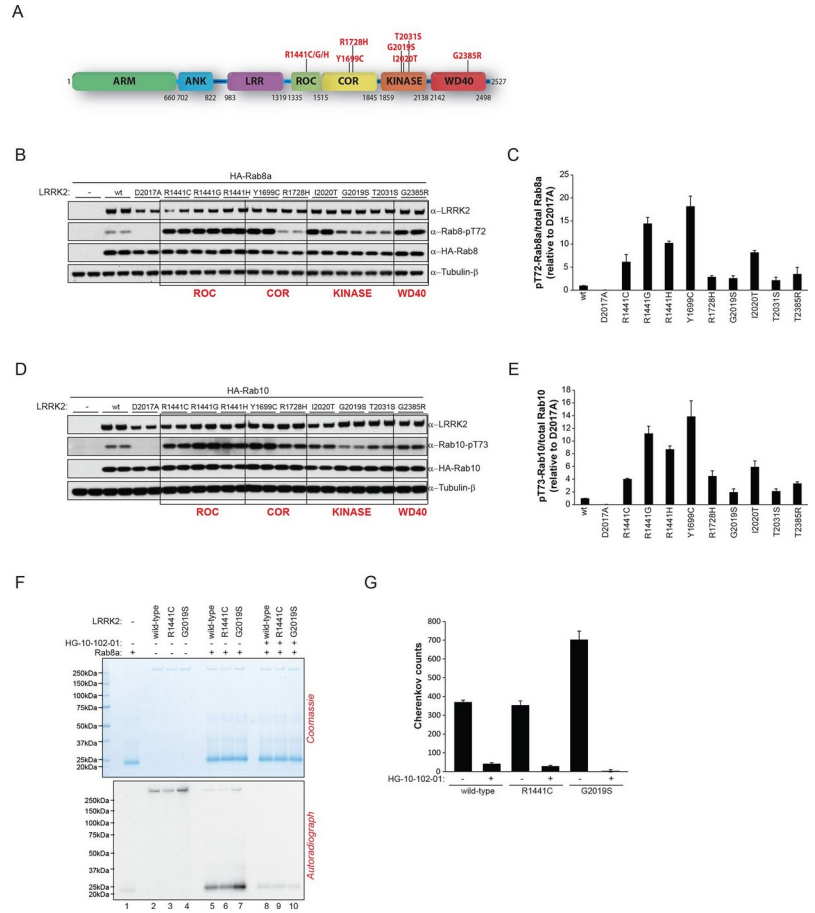
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Pathogenic LRRK2 variants increase Rab phosphorylation

The researchers compared LRRK2 variants associated with Parkinson's disease. Several variants increased phosphorylation of Rab substrates, even though the altered amino acids lay in different functional regions of LRRK2. Their effects converged on the measured kinase output.

The phosphate was added at a conserved position within the Rab switch region. This location is important because other proteins recognize Rab's conformation there. A modification at that site can therefore affect interactions needed for normal trafficking regulation.

The experiments examined both cellular and biochemical conditions. Direct phosphorylation in a defined system helped establish that LRRK2 could act on the substrate itself, while cellular measurements showed that the modification also occurred in a more complex preparation.



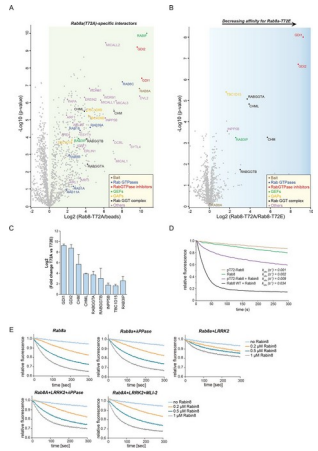
Steger et al. (2016), Fig. 4.

Modified Rabs bind regulatory partners differently

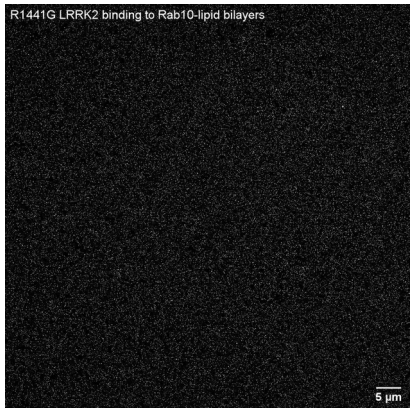
Rab GDP-dissociation inhibitors help regulate the distribution of Rab proteins between membranes and the soluble cellular pool. The LRRK2 study found that phosphorylated Rab proteins had reduced affinity for these regulators. Affinity describes how strongly the partners bind.

A reduction in binding changes access to a normal trafficking-control step. The study therefore supplied a molecular action between increased kinase activity and altered Rab handling. It did not require assuming that all cellular membranes responded identically.

A separate membrane experiment places Rab10 on an artificial lipid bilayer and follows fluorescent LRRK2 binding near its surface. The recording uses the disease-associated R1441G form of LRRK2. Increasing fluorescence at the membrane reveals protein recruitment without requiring an intact neuron or an axonal transport task.



Steger et al. (2016), Fig. 5.



R1441G LRRK2 binding to Rab10 membranes. Vides et al. (2022).

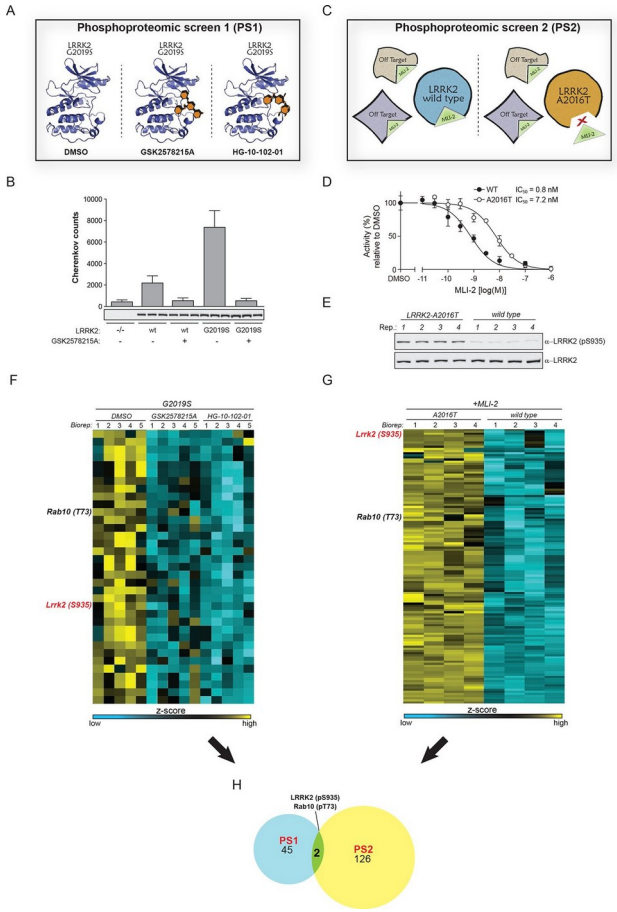
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Deep brain stimulation modifies circuit activity

Deep brain stimulation delivers electrical pulses through implanted electrodes. In Parkinson's disease, targets commonly include the subthalamic nucleus or the internal globus pallidus. Stimulation changes activity within a circuit rather than supplying new dopaminergic neurons.

The effects depend on electrode location, pulse parameters, and the pathways recruited. Describing stimulation only as turning a nucleus off misses the axons and distributed networks it can influence. Its clinical goal is to improve selected symptoms and daily function.

A kinase inhibitor and a stimulation device operate at different biological levels. The former changes a molecular reaction; the latter changes electrical activity in an anatomical network. Comparing their outcomes requires identifying whether a study measures molecular function, motor performance, or progression.



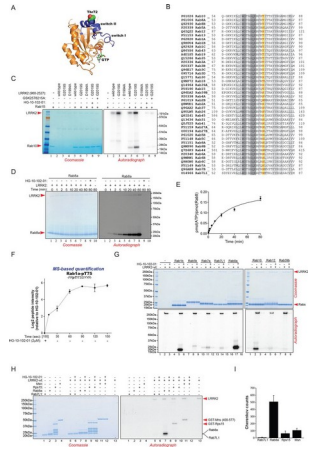
Steger et al. (2016), Fig. 1.

Kinase inhibition tests whether a modification is reversible

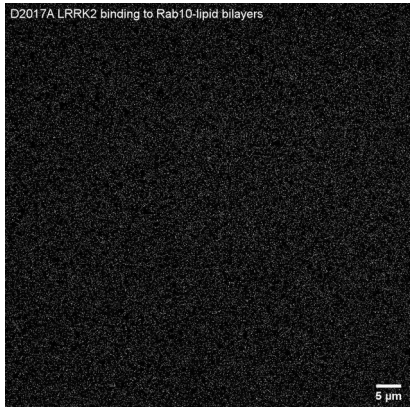
A selective kinase inhibitor reduces an enzyme's phosphate-transfer activity. In the LRRK2 experiments, pharmacological inhibition helped identify phosphorylation events dependent on LRRK2. A change that follows inhibition supplies a temporal test alongside genetic comparisons.

The investigators combined this approach with substrate measurements and biochemical experiments. The convergence identified Rab proteins as genuine targets rather than proteins whose abundance happened to differ between samples. Protein abundance and phosphorylation state were measured as different properties.

The paired bilayer preparation uses D2017A LRRK2, a kinase-inactive form, and follows its binding to Rab10-bearing membranes. This condition separates membrane recruitment from an enzyme's ability to transfer phosphate. It is a different manipulation from adding the kinase inhibitor used in the adjacent experiment.



Steger et al. (2016), Fig. 2.



Kinase-inactive LRRK2 binding to Rab10 membranes. Vides et al. (2022).

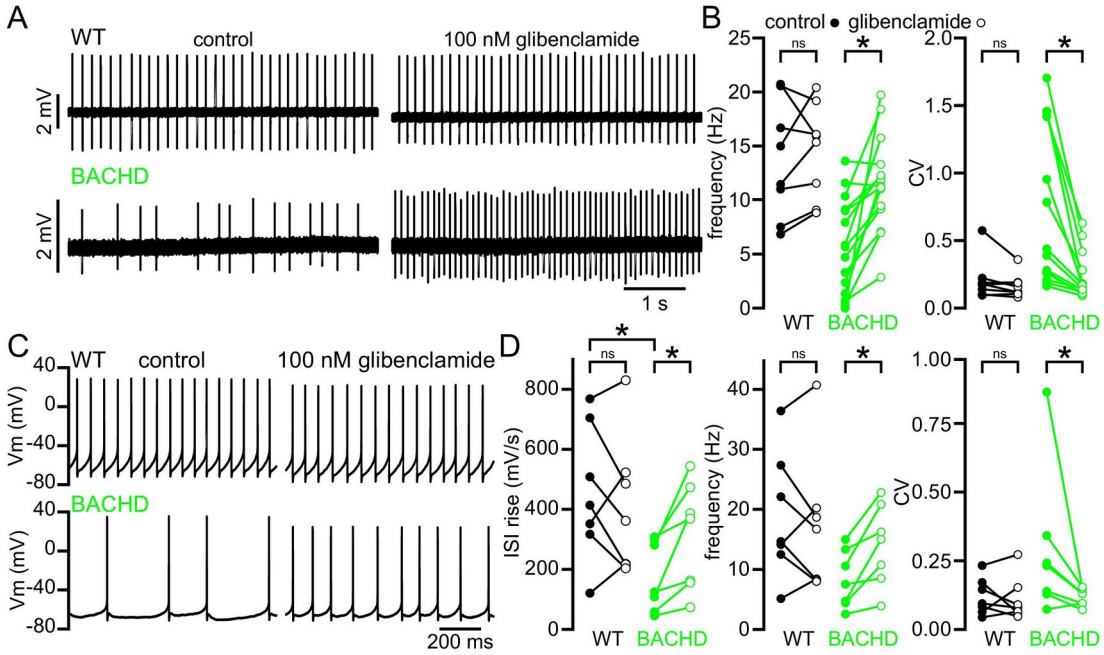
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Huntington's disease changes movement and cognition

Huntington's disease is caused by an expanded CAG repeat in the HTT gene. CAG encodes glutamine, so the expansion produces an abnormally long glutamine segment in huntingtin. The altered protein affects several cellular processes over the course of disease.

The disorder can produce involuntary movements, difficulties with voluntary movement, cognitive changes, and psychiatric symptoms. Striatal medium spiny neurons are especially vulnerable, but the affected network extends beyond the striatum. Symptoms change as the disease progresses.

The repeat expansion follows an autosomal-dominant inheritance pattern. Inheriting a disease-causing expansion differs from inheriting one small statistical risk factor. Genetic counseling therefore addresses both the biological result and the person's choices about learning inherited information.



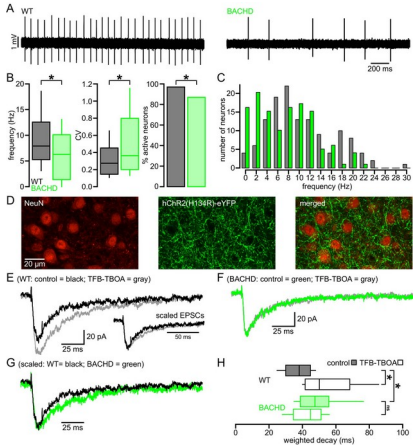
Atherton et al. (2016), Fig. 5.

Subthalamic dysfunction precedes cell loss

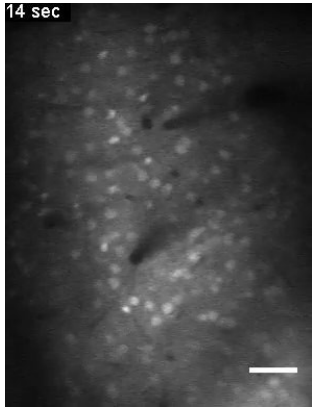
The subthalamic nucleus helps regulate actions within basal-ganglia circuits. Researchers examined this region in two mouse models of Huntington's disease, including a knock-in model carrying an expanded repeat within the relevant genetic context.

At relatively early ages, subthalamic neurons already showed impaired autonomous firing. Autonomous firing is activity generated by the neuron's own membrane properties without requiring each spike to be triggered by a new synaptic input.

Neuronal activity can also be measured through calcium-sensitive fluorescence. Separate imaging in mouse visual cortex follows brief signals in individual cell bodies. These cortical calcium transients are a different measurement from the subthalamic electrical firing changes, although both examine living neurons before counting cell loss.



Atherton et al. (2016), Fig. 1.



Calcium signals in mouse visual cortex. Arnoux et al. (2018).

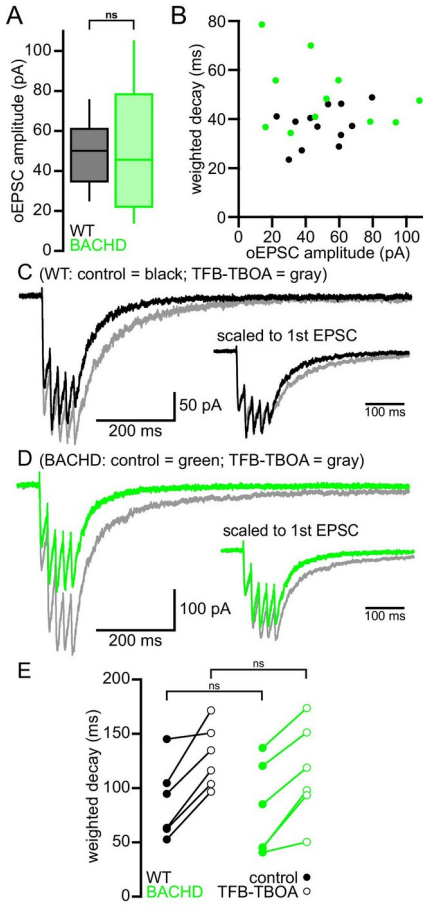
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Prolonged glutamate currents increase the cellular burden

Glutamate transporters normally remove released glutamate from extracellular space. In the Huntington's models, reduced glutamate uptake prolonged NMDA-receptor-mediated currents in subthalamic neurons. The receptor therefore remained activated longer after glutamate reached the synapse.

NMDA receptors allow charged particles, including calcium, to enter under suitable conditions. Prolonged receptor activity changes both electrical excitation and intracellular calcium handling. These effects can increase the burden placed on mitochondria and other cellular systems.

The researchers detected increased mitochondrial oxidant stress alongside the altered synaptic currents. Oxidant stress describes excessive reactive chemical activity that can damage cellular components. It supplied a measurable intermediate between abnormal synaptic input and disrupted neuronal firing.



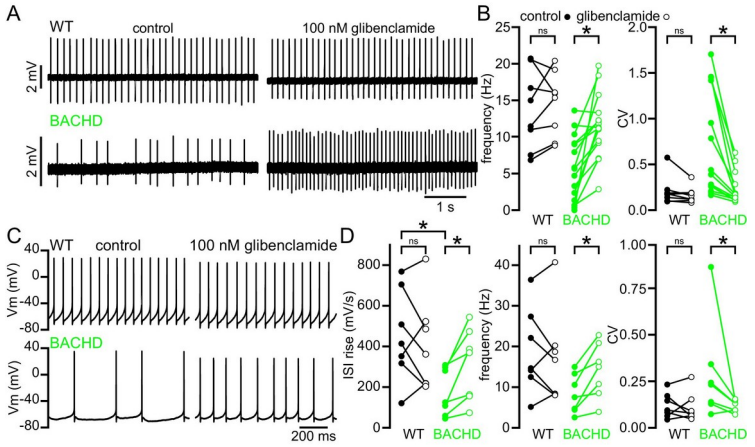
Atherton et al. (2016), Fig. 2.

Expanded huntingtin affects more than visible inclusions

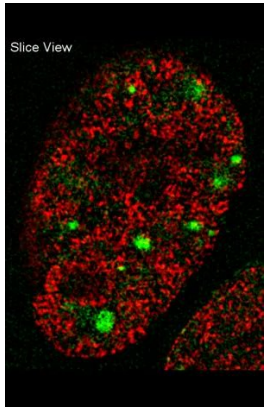
Huntingtin is normally present in cells and participates in processes including transport and cellular organization. An expanded glutamine segment changes its behavior. Soluble abnormal forms, fragments, and larger inclusions need not have identical effects on neuronal survival.

An inclusion is a visible accumulation of material within a cell. Its presence does not by itself establish which molecular species is toxic. A neuron can be affected by altered interactions and transport before a large deposit becomes visible.

Expanded huntingtin can be localized within a cell using fluorescent tags. Three-dimensional imaging of embryonic stem cells places a 94-glutamine huntingtin fragment beside labeled chromatin, the DNA-protein material of the nucleus. Separate colors distinguish the expanded protein from the surrounding nuclear material.



Atherton et al. (2016), Fig. 5.



Expanded huntingtin beside labeled chromatin. Li et al. (2016).

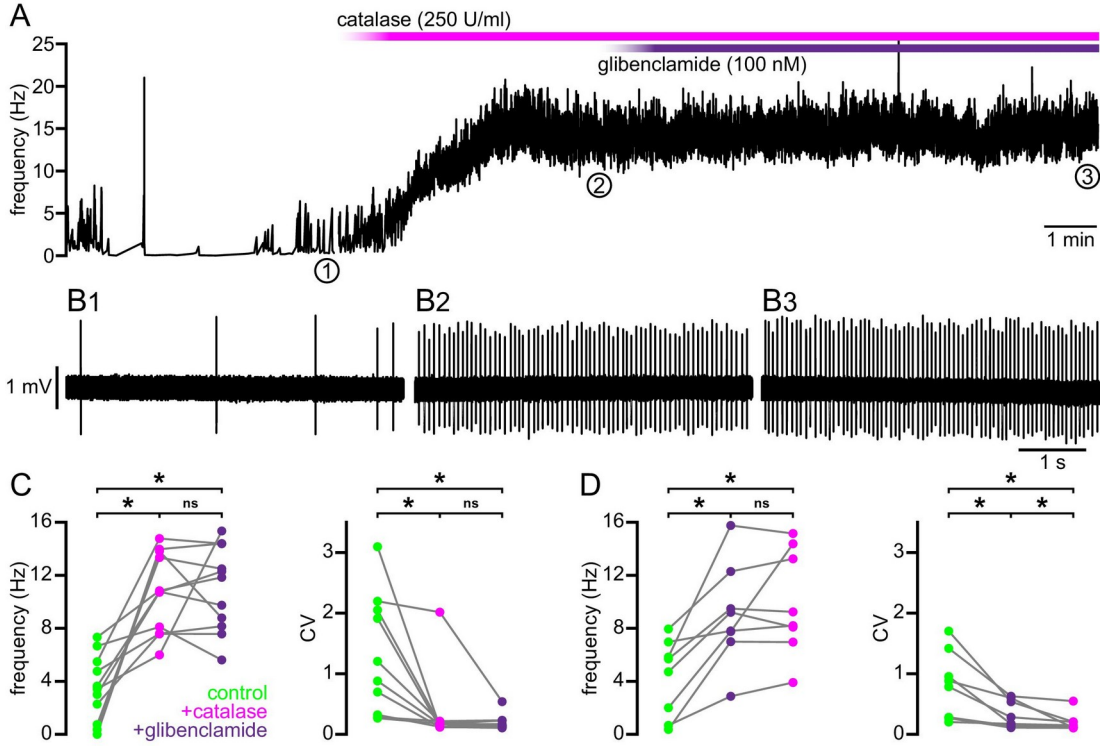
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Blocking the stressed pathway restores firing in slices

The researchers tested whether the subthalamic firing abnormality could be reversed experimentally. NMDA-receptor antagonism reduced the receptor-driven burden, and breaking down hydrogen peroxide reduced oxidant stress. Both manipulations improved the impaired firing pattern in their preparations.

ATP-sensitive potassium channels contributed to the reduced autonomous activity. When these channels open, potassium conductance can hold the membrane away from firing. Their activation provided an electrical route through which the stressed cellular state changed spike production.

These results connected synaptic input, oxidative chemistry, ion-channel activity, and firing. The rescued outcome was neuronal activity in the mouse-model preparation, rather than restoration of every movement or cognitive function affected by Huntington's disease.



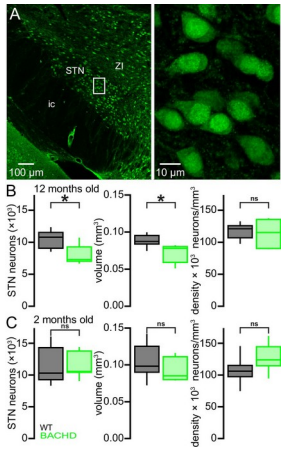
Atherton et al. (2016), Fig. 8.

Later degeneration follows an earlier functional disturbance

The study compared different ages instead of measuring only a terminal disease stage. Subthalamic firing was abnormal before substantial neuronal loss. At a later age, fewer subthalamic neurons remained in the disease-model tissue. Electrical recordings measured the activity of surviving cells, while anatomical counts measured how many neurons remained.

This ordering matters because an abnormal signal can arise from malfunctioning cells that are still alive. Later, the absence of cells adds a different constraint: changing ion-channel activity cannot make a lost neuron participate again. Ion channels are membrane proteins that control the flow of charged particles, allowing a living neuron to generate electrical signals.

A separate Huntington's study uses a trained mouse's visual discrimination behavior as a functional measurement. The animal moves through the apparatus and responds to a visual choice. Such task performance measures an organized action, whereas the subthalamic recordings measure the electrical properties of individual neurons.



Atherton et al. (2016), Fig. 11.



Trained mouse visual discrimination task. Arnoux et al. (2018).

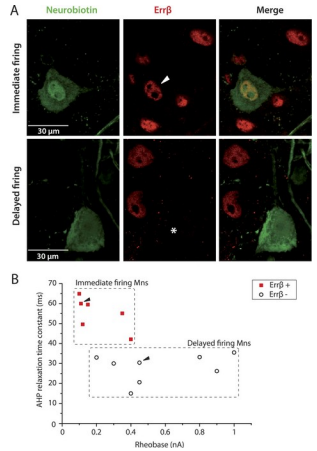
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ALS weakens the pathways that activate muscle

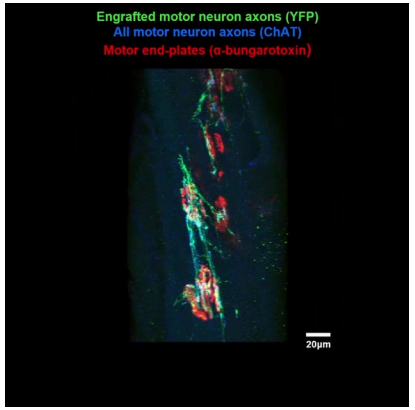
Amyotrophic lateral sclerosis, or ALS, causes progressive degeneration of motor neurons. Upper motor neurons influence spinal and brainstem motor pathways, while lower motor neurons send axons directly to muscle. Damage at these levels produces different combinations of weakness and abnormal muscle control.

A motor unit consists of a motor neuron and the muscle fibers it innervates. Loss of a motor neuron deprives those fibers of neural activation. Surviving units can partly compensate through sprouting, but continued degeneration eventually exceeds that compensation.

A separate ALS repair experiment grafted light-sensitive motor neurons into SOD1-mutant mice. Three-dimensional microscopy shows graft-derived axons reaching muscle endplates, where nerves normally activate muscle fibers. The labeled contact provides an anatomical measure of reinnervation, distinct from the spontaneous progression of untreated disease.



Leroy et al. (2014), Fig. 3.



Grafted motor axons at muscle endplates. Bryson et al. (2024).

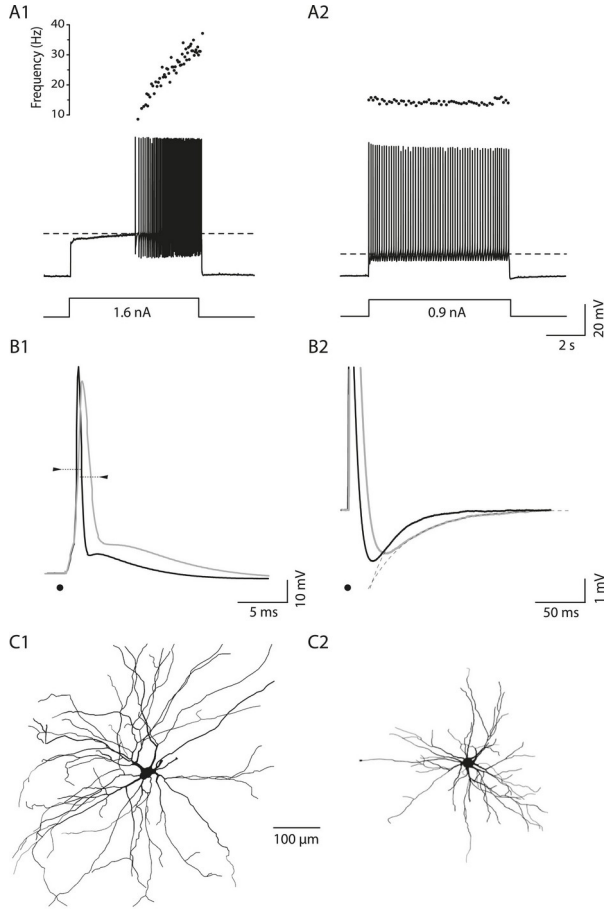
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Vulnerable and resistant motor neurons can be compared

Large motor neurons that drive fast-contracting muscle fibers are especially vulnerable in several ALS models. Smaller motor neurons supplying slow-contracting fibers can be relatively resistant. Researchers recorded from identified motor-neuron types in neonatal mice carrying mutant SOD1.

One proposed explanation was that early excessive excitability selectively harmed the neurons that would later degenerate. Excitability describes how readily a neuron produces action potentials in response to current. The experiment tested this property in the different cell types.

Identifying the motor-neuron type was essential. Averaging every recorded cell together could conceal a difference between a vulnerable population and a resistant one, producing a misleading description of which early property predicted later degeneration.



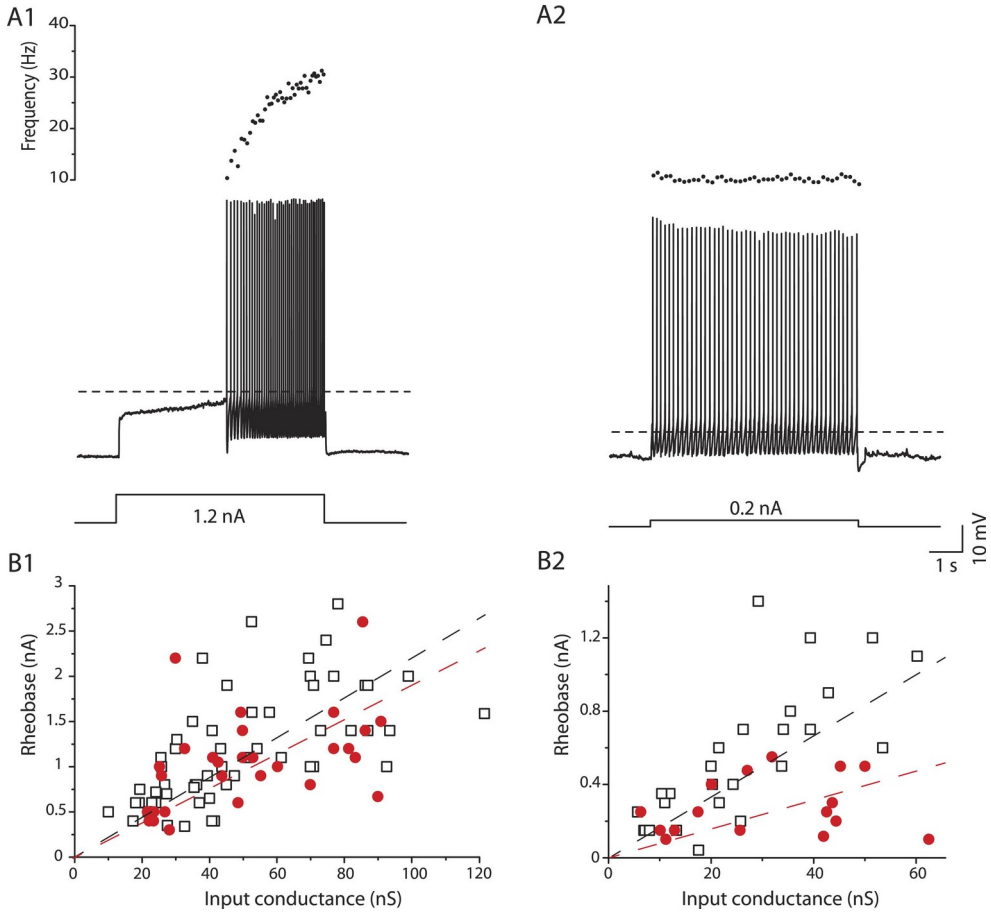
Leroy et al. (2014), Fig. 1.

The resistant neurons were the more excitable population

The vulnerable fast-type motor neurons did not show the predicted increase in intrinsic excitability at the neonatal stage examined. Instead, the resistant slow-type neurons were more excitable in the mutant mice. This reversed the proposed correspondence between early hyperexcitability and vulnerability.

The researchers measured properties such as the current needed to elicit firing and the voltage threshold for a spike. Lower required current can indicate greater excitability. These measurements describe a cell's electrical response, rather than directly counting its later survival.

The result argues against that specific early-hyperexcitability explanation in the tested model and age. It leaves the recorded motor-neuron populations with different physiological changes, even though the most vulnerable population did not carry the predicted early abnormality.



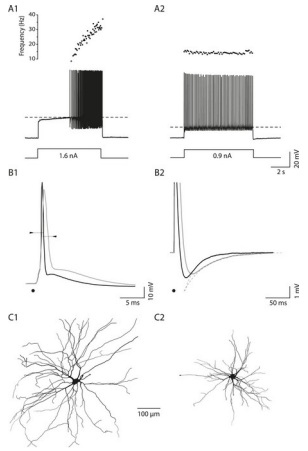
Leroy et al. (2014), Fig. 6.

Motor-neuron loss disconnects muscle from its command

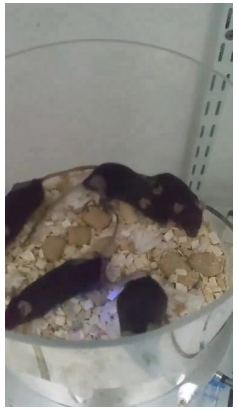
The neuromuscular junction is the synapse between a motor axon and a muscle fiber. Acetylcholine released from the nerve terminal activates muscle receptors, leading to a muscle action potential and contraction. Loss of the axon interrupts this sequence.

Weakness can therefore develop without the muscle initially being the primary site of disease. As neural input disappears, muscle fibers lose effective activation and undergo secondary changes. The affected person's intended action and the remaining motor output become increasingly mismatched.

Light can activate grafted motor neurons engineered to carry channelrhodopsin, a light-sensitive ion channel. The ALS mouse recording shows delivery of this optical stimulation during training. The associated experiment measured muscle force separately, testing whether repeated activation helped the newly supplied motor pathway drive contraction.



Leroy et al. (2014), Fig. 1.



Optical stimulation of grafted motor neurons. Bryson et al. (2024).

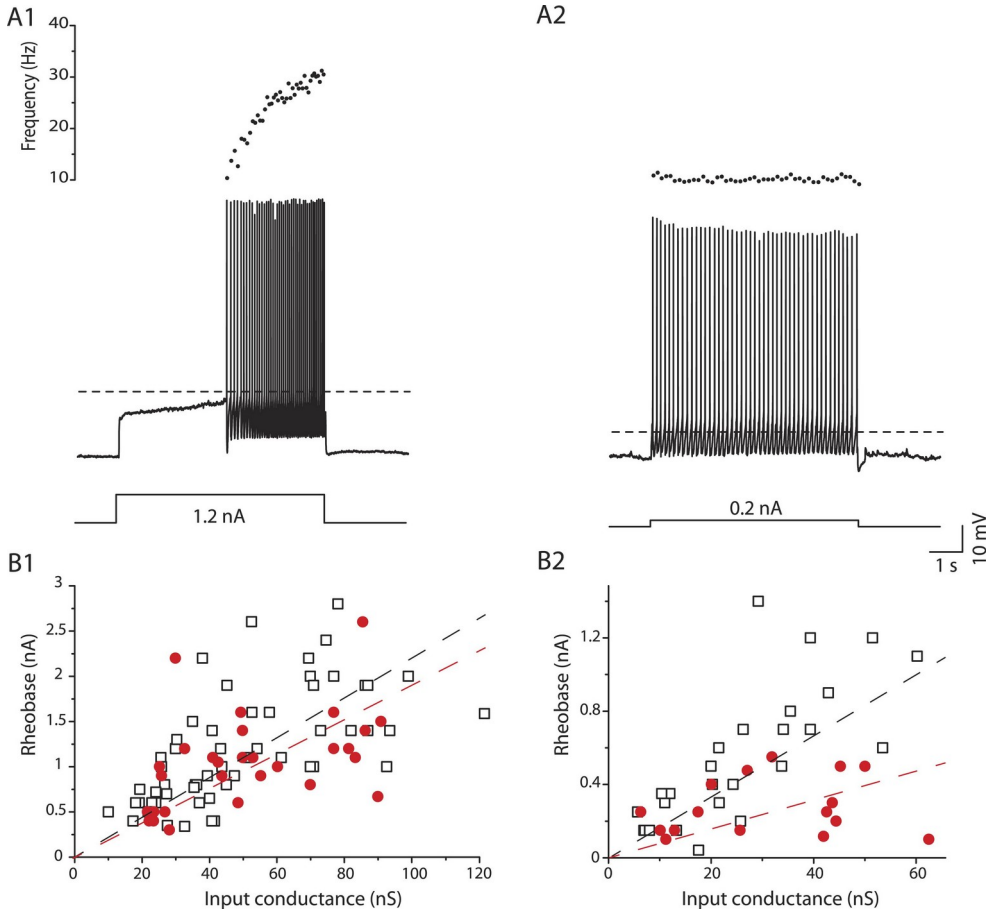
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A hypothesis must predict the vulnerable cell type

If a cellular property explains selective vulnerability, the property should be examined in the cells that actually become vulnerable. The ALS experiment compared fast and slow motor-neuron populations explicitly, rather than inferring vulnerability from a whole-spinal-cord average.

The dissociation was biological: the relatively resistant population showed greater early excitability. The investigators therefore could not use neonatal intrinsic hyperexcitability as a simple explanation for why fast motor neurons later died in this model.

A population's firing response and its degenerative fate are separate measurements. Recording them in the appropriate cell types prevents an early physiological difference from being mistaken for the cause of later cell loss solely because both occur in the same disease model.



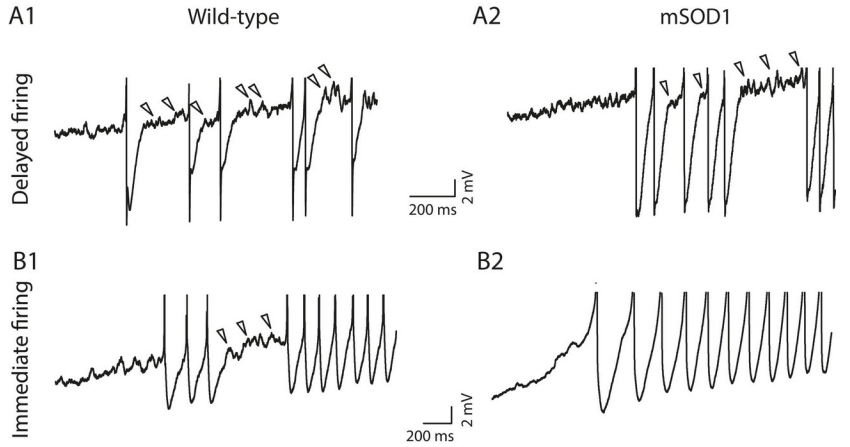
Leroy et al. (2014), Fig. 6.

An early electrical measurement is not a survival assay

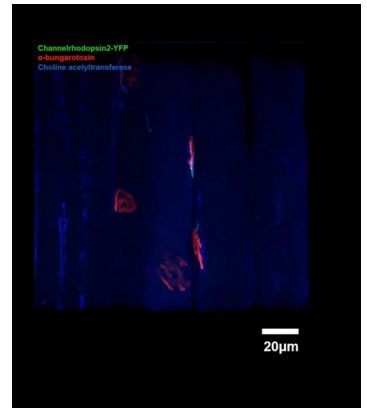
A neuron can change its threshold, membrane resistance, or firing pattern while remaining viable. Electrophysiological recordings reveal those properties at the time of the experiment. Degeneration requires additional evidence about loss of cellular integrity or survival across time.

The mutant-SOD1 study focused on an early developmental stage when physiological abnormalities were already measurable. Its cell-type comparison showed that those abnormalities did not align with later vulnerability in the predicted way. The age and identity of the recorded cells were therefore part of the conclusion.

Muscle anatomy provides another outcome after motor-neuron grafting. A three-dimensional tissue recording follows a longitudinal muscle section and identifies individual fibers for analysis. Fiber structure is measured at a different biological level from the neonatal electrical recordings and requires its own preparation and measurement.



Leroy et al. (2014), Fig. 7.



Muscle-fiber anatomy after motor-neuron grafting. Bryson et al. (2024).

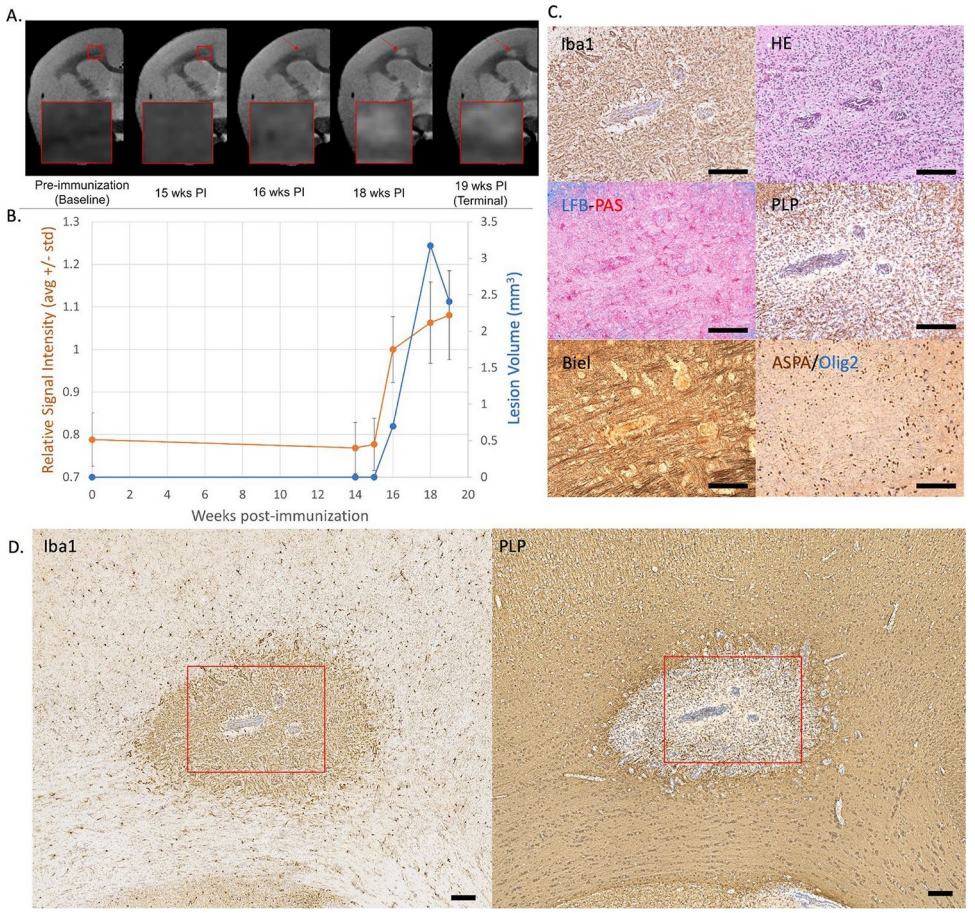
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Multiple sclerosis disrupts insulation and axonal function

Multiple sclerosis involves immune-mediated damage within the central nervous system. Myelin, the multilayered membrane made by oligodendrocytes, normally allows rapid and reliable signaling along many axons. Damage to this insulation changes conduction and can expose axons to further injury.

Demyelination means loss of myelin. Remyelination means formation of new myelin around an axon that has lost it. These processes can occur within different lesions or at different stages, so a single brain can contain tissue undergoing damage and tissue undergoing repair.

Symptoms depend on lesion location and can include visual, sensory, movement, and other neurological changes. A diagnosis does not specify one universal symptom sequence. Relapsing inflammatory activity and accumulating disability also describe different dimensions of the disease course.



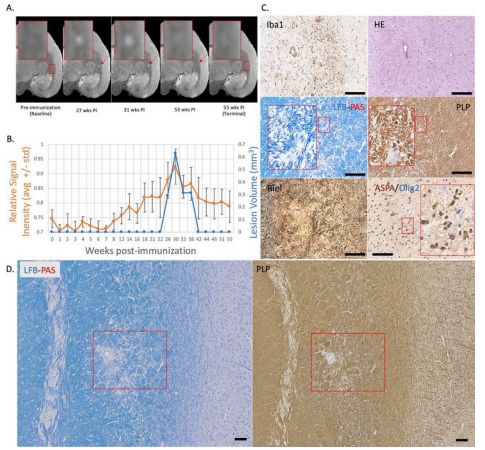
Donadieu et al. (2023), Fig. 1.

MRI can be tested against actual remyelination

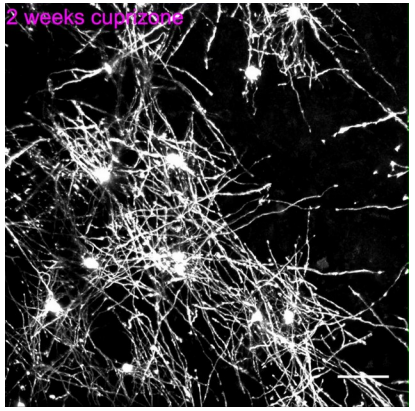
Researchers examined magnetic resonance imaging in a nonhuman-primate model of multiple sclerosis. They compared imaging measurements with tissue evidence of myelin status. This tested whether a signal change corresponded to remyelination rather than simply labeling every brighter or darker region as repair.

Myelin changes the physical environment surrounding water and affects several magnetic resonance measurements. Inflammation, tissue damage, and other processes can also change those measurements. Comparing imaging with histology helps identify what an imaging feature actually represents.

A separate mouse study repeatedly imaged oligodendrocytes, the cells that make central myelin. Cuprizone treatment removed cells and myelin, followed by new cells during recovery. The same cortical volume was followed through the sequence, providing a direct cellular record alongside the separate MRI approach.



Donadieu et al. (2023), Fig. 3.



Oligodendrocyte loss and replacement. Orthmann-Murphy et al. (2020).

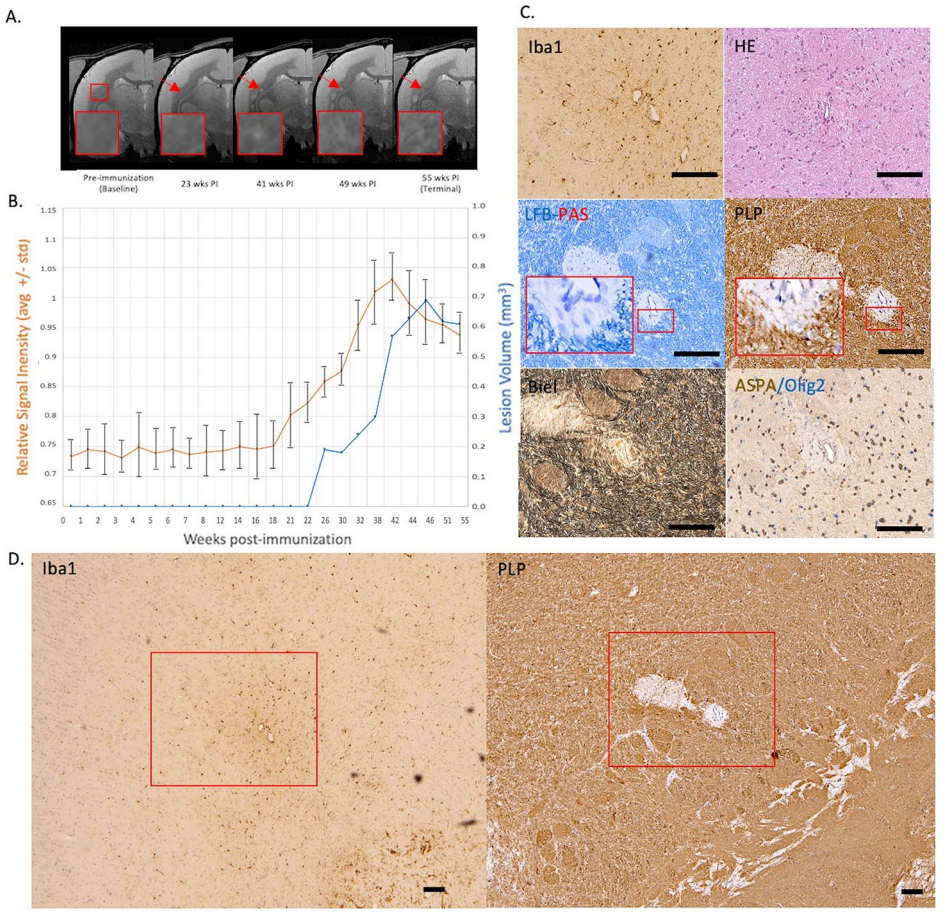
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Repair requires new membrane around surviving axons

Oligodendrocyte-lineage cells can generate replacement myelin under suitable conditions. A surviving axon supplies a different substrate for repair from an axon that has already degenerated. Restoring insulation therefore depends on both the myelin-forming cells and the condition of the axon.

A new sheath may differ in thickness or length from the original sheath. Histological examination can reveal these structural features. An imaging measure that changes during repair needs to be evaluated against such physical evidence of myelin formation.

Remyelination and suppression of inflammation are different treatment goals. Reducing immune attack can protect tissue from new damage, while promoting repair acts on existing demyelinated regions. Their effects can overlap without being the same biological process.



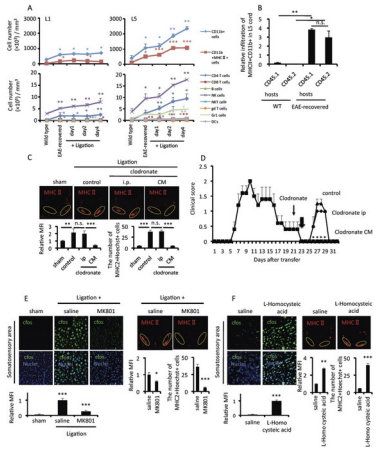
Donadieu et al. (2023), Fig. 2.

Vessel-localized immune entry can amplify a model relapse

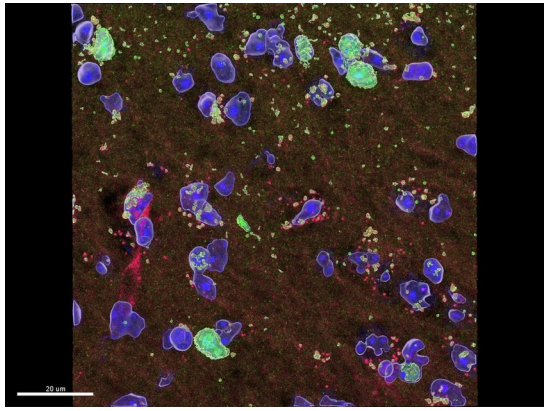
In experimental autoimmune encephalomyelitis, researchers studied how a sensory signal could influence recurrent inflammation. The model produces immune-mediated central nervous system disease in mice. It is not identical to human multiple sclerosis, but allows controlled manipulation of neural and immune pathways.

Pain-related activation recruited a sensory–sympathetic pathway and altered immune-cell accumulation at particular spinal vessels. Chemokines, small proteins that guide cell movement, helped organize recruitment. Antigen-presenting cells accumulated before additional pathogenic immune cells entered the spinal cord.

Immune responses also involve cells inside brain tissue. In a separate mouse EAE preparation, a three-dimensional scan identifies oligodendroglial cells expressing Cd74, part of the machinery that displays protein fragments to immune cells. Olig2 labels their cell lineage, while another color marks Cd74.



Arima et al. (2015), Fig. 4.



Immune-associated oligodendroglia in an EAE mouse. Harrington et al. (2023).

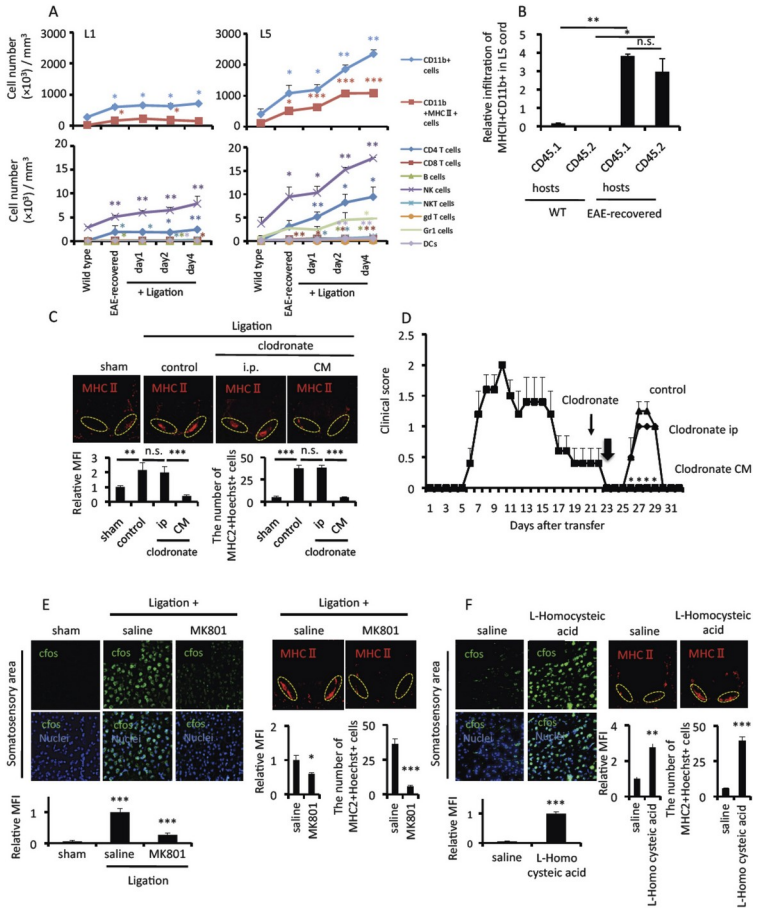
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Inflammation and conduction failure are different events

An inflammatory lesion contains immune activity that can damage myelin and other tissue components. Conduction failure is the inability of an axon to transmit a signal normally. Inflammation can cause or worsen conduction problems, but the two terms describe different processes.

Myelin loss changes current flow along the axon. Damage to the axon itself adds a further problem because replacing insulation cannot restore an axon that no longer remains intact. Clinical changes can therefore reflect several overlapping forms of tissue injury.

MRI, immune measurements, electrophysiology, and functional assessment sample different parts of this process. Evaluating repair requires asking whether new myelin formed, whether axons survived, and whether signaling and behavior improved, instead of treating one measurement as all three outcomes.



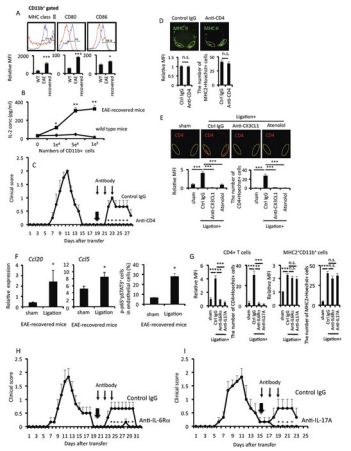
Arima et al. (2015), Fig. 4.

A chemokine pathway directs immune-cell recruitment

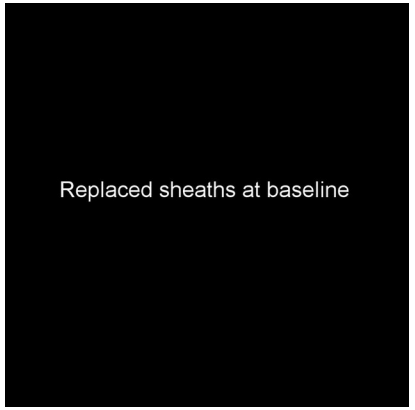
The mouse relapse experiments identified endothelial signaling at specific spinal vessels. Endothelial cells line the blood vessel and regulate interactions between circulating cells and surrounding tissue. Their response helped recruit inflammatory cells into a particular anatomical region.

Blocking or altering steps in the sensory–sympathetic and chemokine pathway changed the relapse response in the experimental system. The intervention therefore tested more than a correlation between discomfort and disease severity: it manipulated components of the proposed recruitment sequence.

Repair can be assessed sheath by sheath after myelin loss. In a separate mouse preparation, three-dimensional imaging marks sheaths that were present before cuprizone, lost, and replaced during recovery. The original blue overlay identifies these structures while the depth scan preserves their positions within cortical tissue.



Arima et al. (2015), Fig. 6.



Individual myelin sheaths lost and replaced. Orthmann-Murphy et al. (2020).

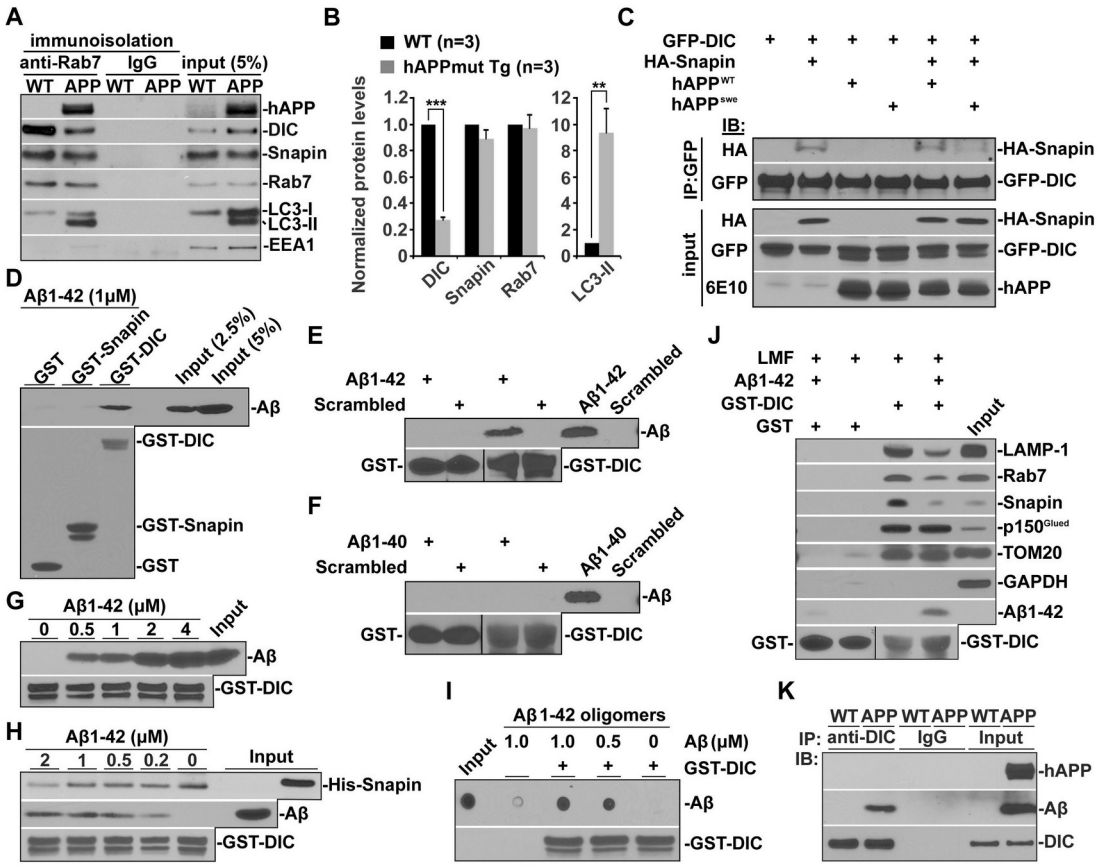
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Dementia is a syndrome with several possible causes

Dementia describes a decline in cognitive abilities that interferes with everyday independence. It is a clinical syndrome rather than the name of a single cellular mechanism. Alzheimer's disease is one cause, while vascular and other degenerative disorders can produce different patterns.

Alzheimer's pathology includes amyloid-associated changes and abnormal tau, alongside synaptic dysfunction and neuronal loss. Memory impairment is common, but attention, language, navigation, and other abilities can also change. The affected network expands as disease progresses.

A biomarker and a person's functional abilities answer different questions. Detecting a molecular pathology identifies a biological feature. Assessing memory, communication, and independence determines how that feature and other processes are affecting daily life.



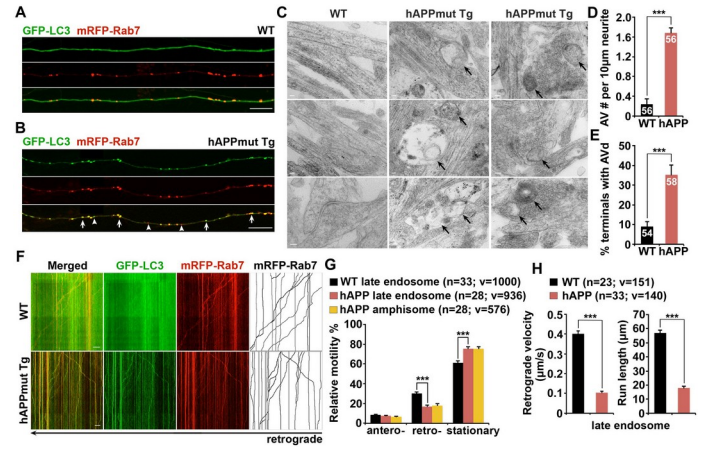
Tammineni et al. (2017), Fig. 4.

Axonal waste must travel toward degradative compartments

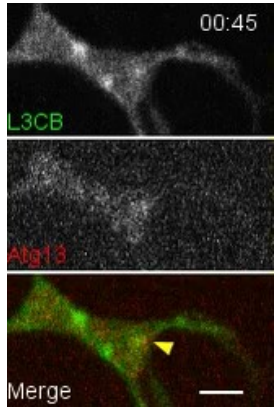
Autophagy packages cellular material into membrane-bound compartments for degradation. In a long axon, newly formed compartments often need to travel toward the cell body, where mature lysosomes are concentrated. This requires attachment to molecular motors moving along microtubules.

Researchers examined autophagic compartments in neurons expressing mutant human amyloid precursor protein and in human Alzheimer's tissue. They found accumulation of amphisomes in distal axons. An amphisome forms when an autophagic compartment merges with an endosomal compartment.

Autophagosomes are membrane compartments that collect cellular material for disposal. In a separate aged-mouse sensory neuron, two fluorescent assembly markers briefly overlap before one leaves the developing compartment. The recording follows productive compartment formation, while the adjacent study examines transport of cargo toward degradative regions.



Tammineni et al. (2017), Fig. 2.



Productive autophagosome formation. Stavoe et al. (2019).

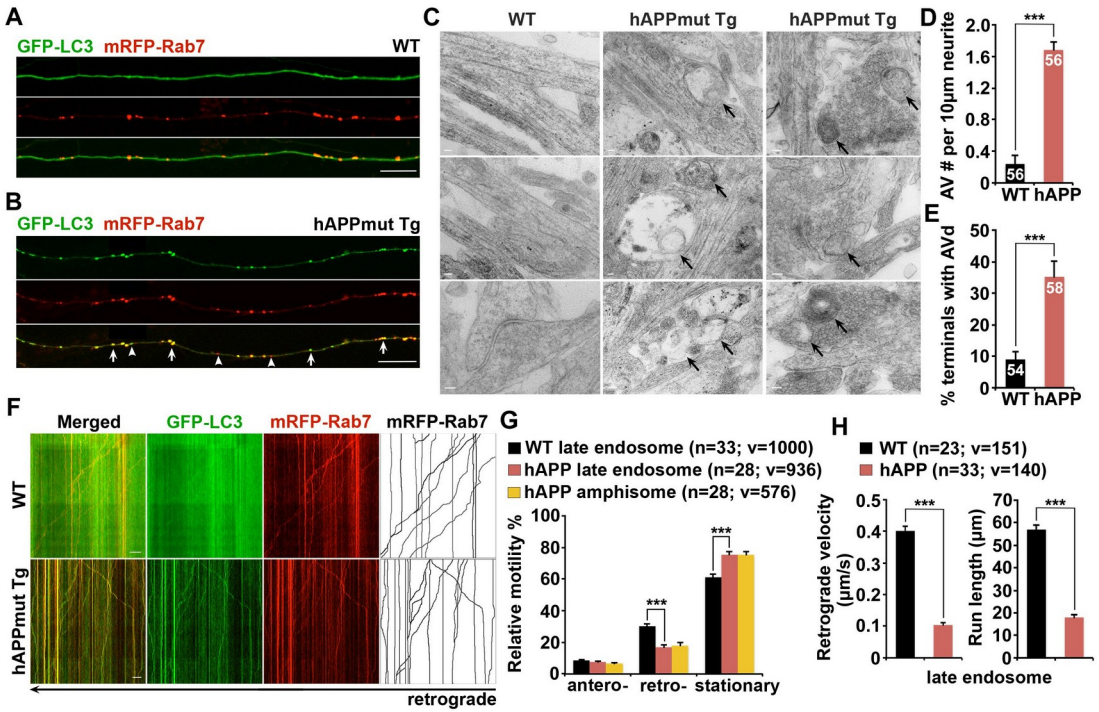
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Amyloid precursor protein can yield several products

Amyloid precursor protein is a membrane-associated protein processed by different enzymes. Certain cleavage sequences release amyloid-beta peptides. The amount, sequence, aggregation state, and clearance of these products influence their effects on surrounding cells and synapses.

Tau normally associates with microtubules, the structures used for intracellular transport. Abnormal tau behavior can disturb this organization and contribute to pathology. Amyloid-related and tau-related processes interact with other cellular systems rather than replacing every aspect of neuronal biology.

A deposit visible in tissue is one stage of molecular accumulation. Smaller assemblies can affect cellular interactions before a large deposit appears. Experiments therefore distinguish soluble forms, deposited material, transport defects, and loss of synaptic function.



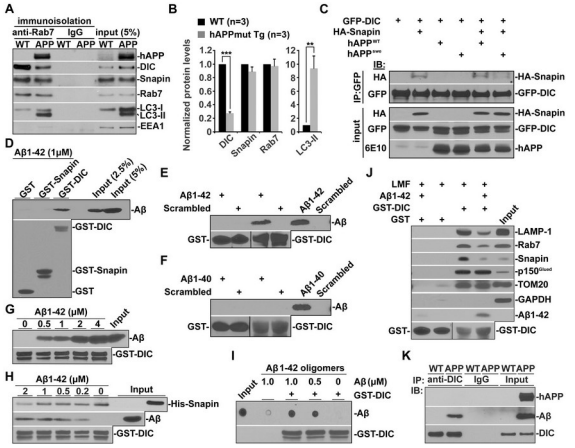
Tammineni et al. (2017), Fig. 2.

Motor-adaptor competition stalls degradative cargo

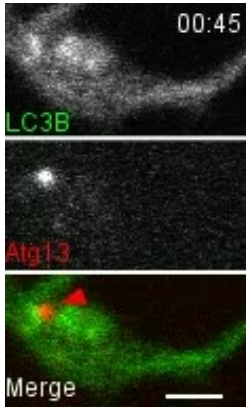
The study found that amyloid-beta oligomers associated with autophagic compartments and interfered with dynein recruitment. Snapin normally helps couple the motor to suitable cargo. Disrupting this motor-adaptor interaction reduced retrograde movement toward the cell body.

Increasing Snapin expression improved retrograde transport in the mutant-APP neurons and reduced accumulation at presynaptic terminals. This intervention changed access to an existing transport mechanism, rather than directly digesting every accumulated compartment at the terminal.

A separate aged-mouse sensory neuron also shows stalled autophagosome formation. The two labeled assembly proteins fail to overlap at the marked sites. This is an assembly defect, distinct from competition for the motor adaptor that moves an already formed compartment along an axon.



Tammineni et al. (2017), Fig. 4.



Stalled autophagosome formation. Stavoe et al. (2019).

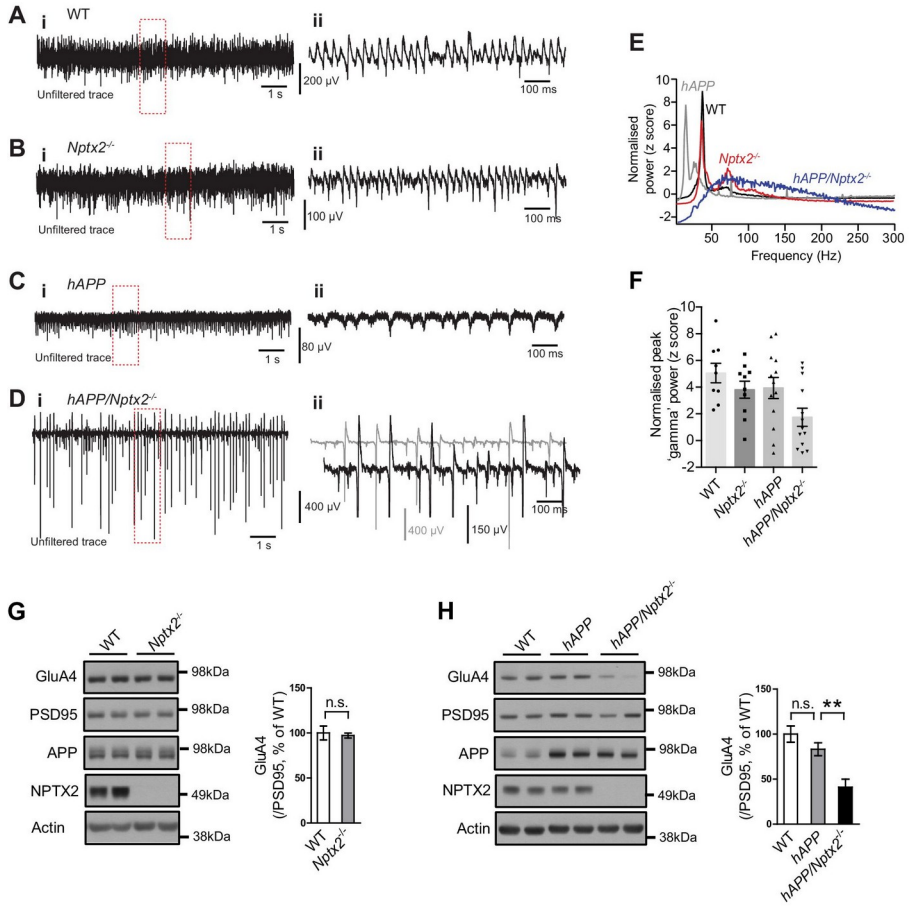
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Interneuron recruitment stabilizes memory networks

Parvalbumin interneurons provide fast inhibition that helps coordinate cortical and hippocampal activity. They need excitatory input from pyramidal neurons to be recruited at the appropriate time. Weakening that input can disturb network timing even when the interneurons remain present.

Researchers studied NPTX2, a protein regulating excitatory synapses onto these interneurons. In an amyloid mouse model, loss of Nptx2 reduced the GluA4 AMPA-receptor subunit, disrupted rhythmic activity, and increased pyramidal-cell excitability. The mechanism involved failed recruitment of inhibition.

Human Alzheimer's cortex also showed reduced NPTX2 and GluA4, and cerebrospinal-fluid measurements varied with cognitive performance. These human observations complemented the experimental manipulation in mice while measuring different preparations and different parts of the proposed pathway.



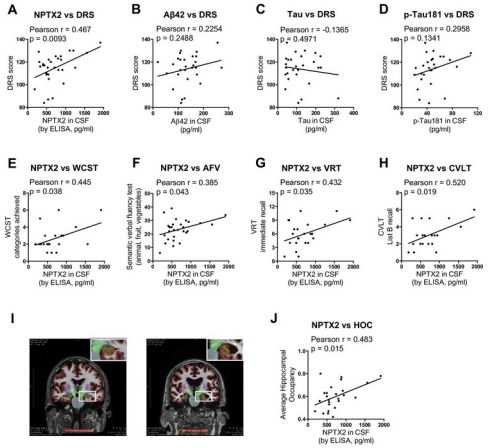
Xiao et al. (2017), Fig. 3.

Memory loss reflects more than one damaged structure

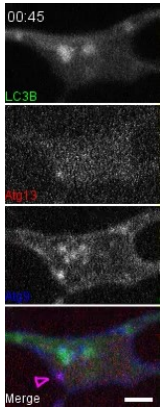
Episodic memory depends on interactions among hippocampal, diencephalic, and cortical regions. Damage to one part can interrupt a larger pathway. A person may therefore have severe difficulty learning new information even when many stored skills and familiar behaviors remain available.

In Alzheimer's disease, synaptic and network dysfunction can precede extensive neuronal loss. In other syndromes, a different anatomical pattern produces a different memory profile. The symptom of forgetting does not by itself identify which molecular process caused it.

Cellular housekeeping can vary even within one neuron. Imaging of a young adult mouse sensory neuron shows both productive and stalled autophagosome formation at different sites. These local events are cellular measurements, separate from the learning and recall tasks used to characterize a person's memory loss.



Xiao et al. (2017), Fig. 6.



Productive and stalled events in one neuron. Stavoe et al. (2019).

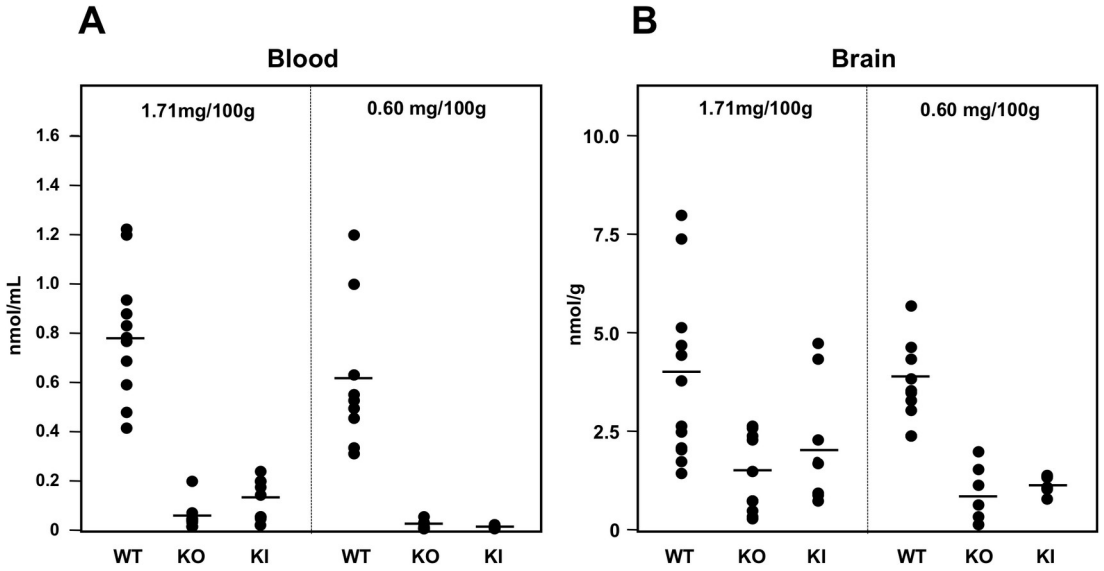
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Thiamine deficiency can damage memory pathways

Thiamine is vitamin B1, a nutrient required for enzymes involved in cellular energy metabolism. Severe deficiency can impair brain function. The nervous system is especially vulnerable because maintaining signaling and ion gradients demands a continuous supply of usable energy.

Wernicke encephalopathy and Korsakoff syndrome are related but distinct clinical descriptions. An acute deficiency state can produce confusion, eye-movement abnormalities, and impaired coordination. Persistent damage can leave severe memory impairment, including difficulty forming new enduring memories.

Heavy alcohol use can contribute through poor nutrition, impaired absorption, and other physiological effects, but thiamine deficiency has additional causes. The mechanism is not simply exposure to alcohol: access to an essential metabolic cofactor becomes inadequate for vulnerable tissue.



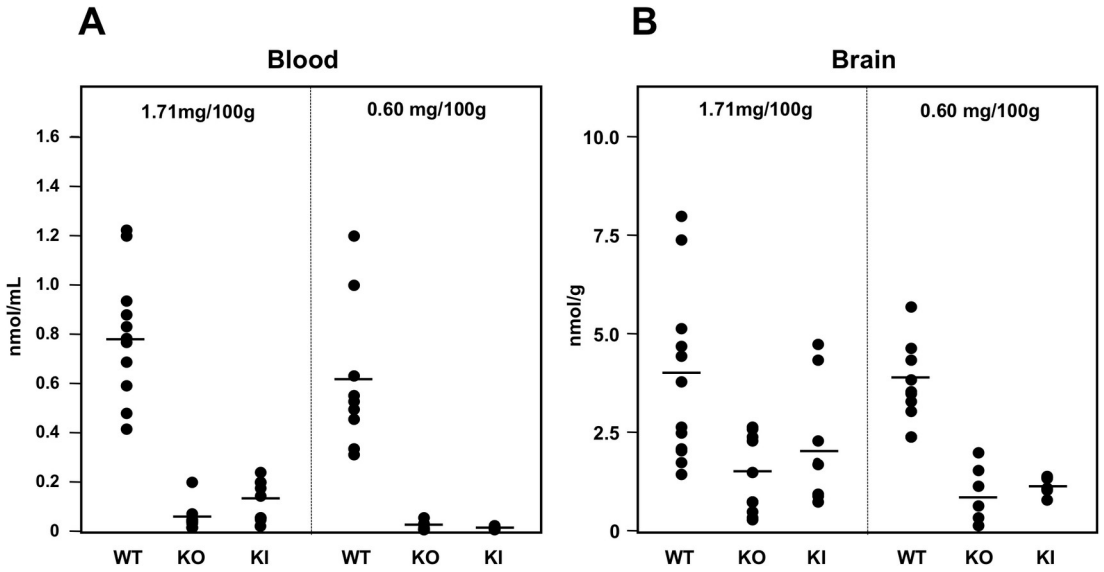
Suzuki et al. (2017), Fig. 2.

A thiamine transporter controls access to the nutrient

The SLC19A3 gene encodes a thiamine transporter. A transporter moves a substance across a cell membrane, so a defect can impair cellular access even when the nutrient exists elsewhere in the body. Researchers examined mice lacking this transporter.

The study compared disease-related outcomes under different thiamine conditions and examined brain lesions and survival. High-dose thiamine prevented brain lesions and prolonged survival in the deficient mice. The intervention changed nutrient availability in a genetically defined transport disorder.

This model is not the same as alcohol-associated Korsakoff syndrome. It isolates a route through which inadequate thiamine access can damage the brain, while allowing a direct test of whether changing vitamin availability alters the experimental outcome.



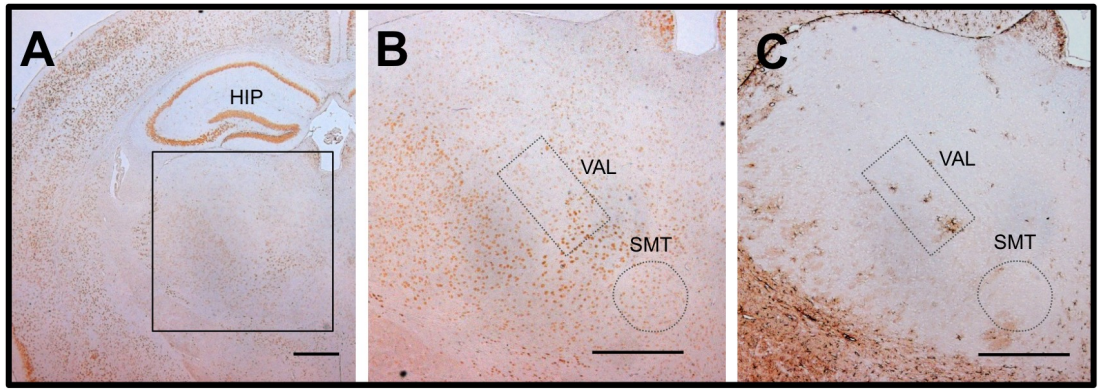
Suzuki et al. (2017), Fig. 2.

Metabolic rescue differs from memory repair

Increasing vitamin availability can change the substrate reaching remaining transport routes or metabolic processes. In the transporter-deficient mouse study, the reported outcomes included prevention of brain lesions and prolonged survival. Those outcomes were measured under the study's defined treatment conditions.

Preventing a lesion differs from rebuilding a memory circuit after neurons have already been lost. The timing of nutrient replacement therefore changes the biological question being tested. Protection during an emerging metabolic disorder and recovery from established amnesia are not interchangeable.

The experimental animals carried a defined genetic defect in thiamine handling. Human nutritional deficiency can involve additional medical conditions and different durations of deprivation. The study supplied a concrete metabolic mechanism without making every thiamine-related syndrome identical.



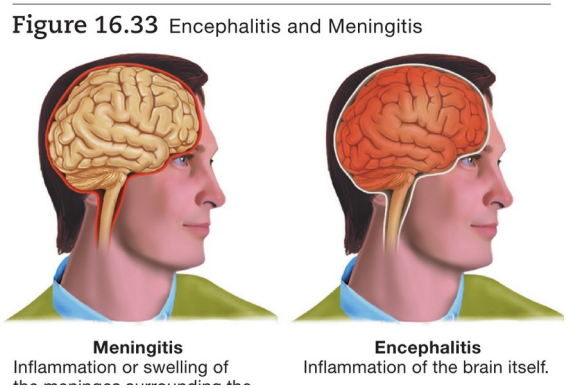
Suzuki et al. (2017), Fig. 8.

Encephalitis directly affects brain tissue

Encephalitis is inflammation involving brain tissue, often associated with infection. The affected regions influence the symptoms, which can include altered awareness, seizures, or changes in memory and behavior. The clinical problem includes both the infectious agent and the host response.

Some viruses enter neural tissue and injure cells directly, while immune responses can add further damage. Treatment depends on the suspected cause and the person's condition. Antiviral treatment for a susceptible virus acts differently from treatment aimed at swelling or seizure control.

Rabies virus can travel inside peripheral axons toward the nervous system. Fluorescent imaging of cultured sensory-neuron axons follows viral particles moving along these narrow processes. This retrograde movement carries cargo toward cell bodies, providing a route different from direct entry across a cerebral blood vessel.



Carlson and Birkett (2022), Fig. 16.33, p.545.



Rabies particles traveling along sensory axons. Gluska et al. (2014).

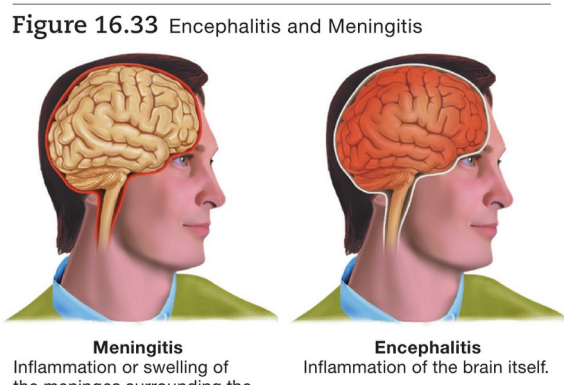
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Meningitis affects the coverings and surrounding fluid

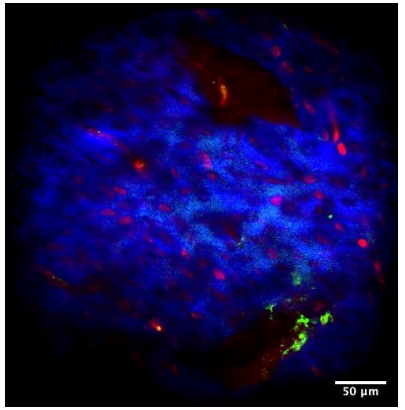
Meningitis is inflammation of the meninges, the membranes surrounding the brain and spinal cord. Infection and inflammation in this compartment can cause headache, neck stiffness, fever, and altered neurological function. Suspected acute bacterial meningitis requires urgent medical assessment.

The affected compartment differs from the brain tissue emphasized in encephalitis. Cerebrospinal-fluid examination can help identify inflammation and its cause. Clinical assessment also considers whether swelling or other abnormalities make a particular diagnostic procedure unsafe.

Fluid around the brain can also enter channels through the skull toward marrow. A depth scan of an isolated mouse skull shows an injected CSF tracer beside labeled channel vessels. This anatomical route links the fluid compartment to nearby marrow rather than depicting bacteria moving through neural tissue.



Carlson and Birkett (2022), Fig. 16.33, p.545.



CSF tracer in skull channels toward marrow. Pulous et al. (2022).

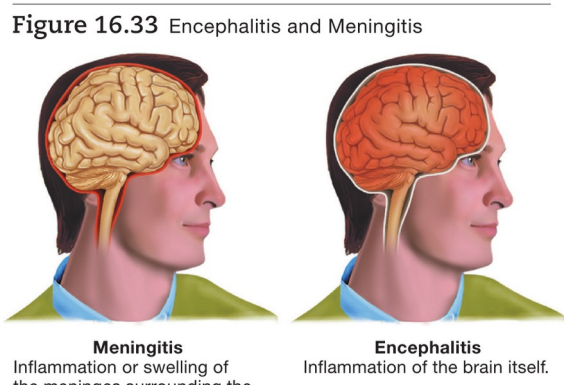
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Infection can reach the nervous system by several routes

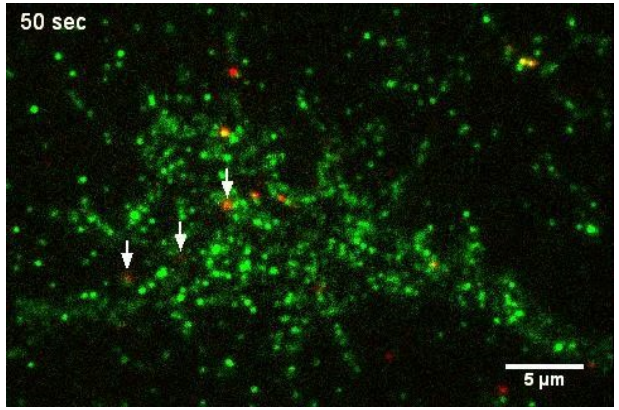
Microorganisms can reach nervous tissue through blood, direct extension from nearby infection, or movement along particular nerves. The route of entry influences which compartments are exposed first. An infection in a covering membrane can subsequently affect surrounding fluid and tissue.

The blood–brain barrier normally restricts movement from blood into the brain. Inflammation can alter this interface and recruit immune cells. Protective immune activity and tissue injury can occur together when the response changes vascular permeability and local cellular function.

Rabies entry begins before long-distance axonal transport. Imaging at sensory-neuron tips shows labeled viral particles entering with the p75 neurotrophin receptor. The two colors identify virus and receptor moving together during uptake, providing a cellular step by which an infectious agent gains access to an axon.



Carlson and Birkett (2022), Fig. 16.33, p.545.



Rabies and p75 receptor uptake at axon tips. Gluska et al. (2014).

Play video

Key takeaways

Location explains the functional deficit. Tumors, seizures, vascular lesions, trauma, and infection disrupt different cells, compartments, and pathways; identify the anatomy before explaining the behavior.

Timing separates distinct treatment targets. Acute rescue of blood flow, control of seizure activity, rehabilitation, and protection from degeneration act at different stages of injury and recovery.

Protein handling depends on concrete machinery. PINK1, LRRK2, huntingtin-associated pathways, and axonal transport influence substrate recognition, phosphorylation, organelle quality, or movement of cellular cargo.

Selective vulnerability requires cell-specific evidence. The cells with the largest physiological change are not necessarily those that later die; compare vulnerable and resistant populations directly.

Repair is a measured biological outcome. New myelin, restored cargo transport, changed firing, and recovered movement are different outcomes and must be assessed separately.

Nutrients and infectious agents have distinct routes. Thiamine access determines metabolic capacity, while encephalitis and meningitis involve different anatomical compartments and cause-specific treatment.